



Faculty of Engineering,
Built Environment and
Information Technology

Increasing the productivity of waste reclamation efforts

Design Project – MOX 410
Final Report

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Submission date: 26/05/2026

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Keep the writing grounded in the information I have provided to you.

Use South African / British spelling where appropriate.

Focus on clarity, technical accuracy, and logical flow.

Keep the meaning as close as possible to my original writing when refining or editing paragraphs.

Executive Summary

This report presents the design of a practical improvement system that can be attached to a waste picker's trolley. Waste pickers regularly transport heavy loads over long distances using modified or improvised trolleys. These trolleys can cause high levels of physical strain and poor control, and pose safety risks. The goal of this project was to find a low-cost solution that could reduce fatigue, improve trolley control, and increase waste pickers' productivity.

Several concepts were generated and compared against a baseline representing the existing trolleys commonly used. The final concept uses a rear bicycle wheel assembly to improve rolling and vibration performance and directional stability. A universal attachment bracket was developed to connect the wheel-mounting arm to a variety of trolley configurations, and a pulling handle with a manual braking system was added to improve ergonomics and control.

The detailed design focused on the wheel mounting arm, universal attachment bracket, and pulling handle. The load-bearing components were designed with a global safety factor of 2, based on an expected maximum load of 250 kg, yielding a design load of 500 kg. The design also prioritised manufacturability, material availability, cost and environmental impact.

The proposed design provides a practical and realistic improvement to the existing waste pickers' trolley. It is expected to reduce rolling resistance and vibrations, improve comfort, increase stability and improve stopping ability. It is recommended that physical testing and field trials be conducted to verify and refine the design under real operating conditions.

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1 Introduction

1.1 Background

South Africa suffers from a high unemployment rate, which has forced many people living in poverty to collect recyclables to earn income. These people collecting the recycling operate in the informal sector; they are often referred to as “waste pickers”. These waste pickers perform an important service by preventing recyclable materials from entering landfills.

Waste pickers often rely on improvised solutions to when collecting heavy recycling loads. These solutions range from modified shopping trolleys to pallets they have found and fitted with caster wheels. These means of transport are cheap to produce, but they can often be difficult to use for long periods or even put waste pickers in danger. The daily life of a waste picker involves navigating busy urban areas and travelling with heavy loads up steep inclines. The strain on waste pickers' bodies can lead to long-term health problems and limit the amount of recyclable material collected.

1.2 Problem statement

The methods currently used by waste pickers in South Africa place significant physical strain on them. This directly limits their overall productivity and can cause long-term health issues. The collection process involves intense manual labour, with physical collection followed by transporting the reclaimed waste through both urban and congested industrial areas, as well as steep gradients, resulting in exhaustion and reduced productivity.

There is a significant need to identify inefficiencies affecting the daily workflow of waste pickers. The goal of this project is to find a solution that reduces the physical strain experienced and/or increases daily income. The aim is to improve their overall quality of life and hopefully provide some financial security.

2 Requirements analysis

2.1 Needs Analysis

Before evaluating specific solutions, it is important to establish a foundation for understanding the waste pickers' needs. For waste pickers, these needs can be categorised into the following groups:

2.1.1 Physical and Ergonomic strain

- **Chronic Fatigue:** To move the large distances that they travel, they typically burn more calories than they can afford to buy the food needed to reclaim them. Over time this leads to physical wasting.
- **Musculoskeletal Injury:** The constant pulling (often only using one side of the body) can result in long-term damage to their shoulders, lower back, and knees due to repetitive stress.
- **Control Issues:** When the waste pickers are travelling on the steep gradients, the risk that the cart will run away is high. Keeping control of the cart's momentum can be challenging due to the heavy load. This creates a dangerous situation and makes moving the cart more exhausting.

2.1.2 Terrain and Environmental Obstacles

- **Rolling resistance:** The caster and small trolley wheels that are often used for the trolleys are intended to be used on flat and smooth retail floors. When these wheels are used on tar roads, gravel, or sand, the rolling resistance and vibration significantly increases.
- **Weather Exposure:** The waste pickers operate in all weather conditions, ranging from extreme heat to heavy rain and electric storms. The rain makes the individual cold, increases the weight of recycling, and makes the road surface slippery and dangerous.
- **Road Safety:** In the early mornings or bad weather, the carts are often hard to see. They can also obstruct traffic, leading to high accident rates.

2.1.3 Logistic and Economic Inefficiency

- **Volume and Weight:** The waste pickers are often limited by the volume that their trolley can hold or the weight that they can move. If either of these limits is reached early in the day, they risk leaving valuable recycling behind.
- **Security of Goods:** When the waste picker leaves the cart unguarded to look for recycling in bins or enter a building, they risk their goods being stolen or removed by municipal authorities.

2.2 Technology Assessment

The Greater Tygerberg Partnership (GTP) has developed new trollies for the waste pickers to use. The goal of the new trollies developed by GTP is to make waste collection safer and more efficient [1]. They have developed an improved trolley with an extended top section, increasing its capacity and allowing the waste picker to collect more per run. GTP has also developed a waste-collection bike that will further increase the efficiency and productivity of waste pickers. The developed waste trolley and waste bike are shown in Figure 1.



Figure 1: Waste Trolleys (left), waste bike (right) [1]

These waste trollies and waste bikes rely on community involvement to provide waste pickers with improved equipment.

Urban Surfer [2] has also developed a new trolley for the waste pickers to use. They have divided their waste trollies into three main categories: the collection trolley (Figure 2, left), collection bags (Figure 2, middle), and the shelter cover (Figure 2, right).



Figure 2: Collection Trolley (left), Collection Bags (middle), Shelter Cover (right) [2]

Urban Surfer has found a solution that enables the waste picker to improve the collection trolley over time. The collection trolley is designed to be lightweight and durable, with reflective surfaces to increase its visibility to motorists. The trollies also come equipped with GPS to increase security, provide data to the community, and to report on brand penetration and coverage to the supporting companies. The collection bags are custom-made with heavy-duty materials to maximise space on the trolley. The shelter cover is a waterproof cover that protects the recycling during rainy weather and also serves as a temporary shelter for the waste picker. Instead of relying on community involvement or government funding, Urban Surfer has enabled companies to

place their logos or brand names on the trolley and its accessories. This will help to gather funding and help as many waste pickers as possible.

2.3 User Requirements

- **R1: The device must either increase efficiency or reduce fatigue,**
 - The device must, in some way, increase efficiency and productivity by either reducing fatigue or increasing load capacity.
- **R2: The device must be affordable,**
 - The device must be possible to purchase within 3-4 months of work.
- **R3: The device must be able to attach to an existing cart,**
 - The waste picker should not have to alter their cart so that they can use the device.
- **R4: The device must be discreet,**
 - The device must not appear as a high-value product to try to reduce the risk of theft.
- **R5: The device must be easy to repair,**
 - It must be possible to repair the device with cheap or salvaged materials (eg. wire, bicycle inner tube, or a common fastener).
- **R6: The device must not require any specialised tools,**
 - It must be possible to install, adjust, and repair each module with only common tools (eg. screw drivers and spanners).
- **R7: The device must be intuitive to operate,**
 - The waste pickers must be able to use the device with limited to no training.
- **R8: The device must be weather-resistant,**
 - The device must be able to last when continuously operating in harsh conditions (eg. rain and dust).
- **R9: The device must be able to handle high loads,**
 - The device must be able to support the weight of the recycling without the risk of failure.
- **R10: The device must not require regular maintenance,**
 - Once the device has been fitted to the cart, it must last a long time before it requires maintenance.

Table 1: Summary of User Requirements

R#	User Requirement
R1	The device must either increase efficiency or reduce fatigue
R2	The device must be affordable
R3	The device must be able to attach to an existing cart
R4	The device must be discreet
R5	The device must be easy to repair
R6	The device must not require any specialised tools
R7	The device must be intuitive to operate
R8	The device must be weather-resistant
R9	The device must be able to handle high loads
R10	The device must not require regular maintenance

3 Functional Analysis

Based on the user's needs and the identified requirements, the functions that the device must include can be categorised into several main groups. These groups include interface with the cart, motion management, withstanding environmental conditions, and servicing and repair. The function tree shown in Figure 3 provides a breakdown for each group.

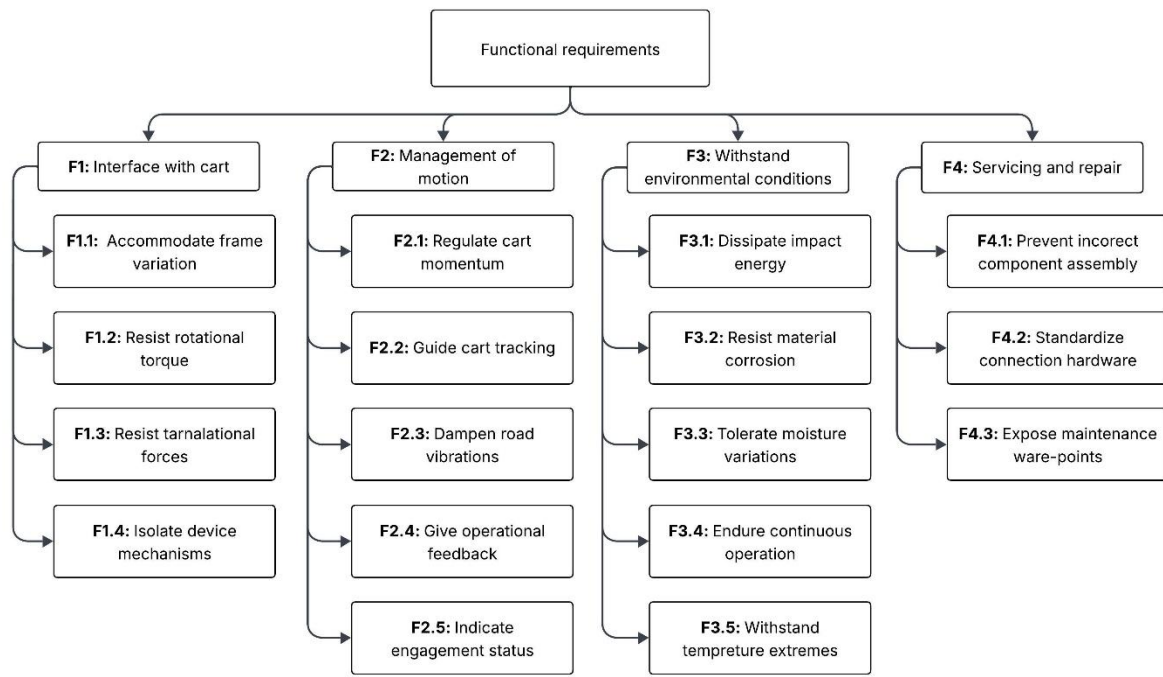


Figure 3: Function Tree

4 Technical Design Specifications (TDS)

The technical design specifications for the device that will help the waste reclaimers are shown in Table 2.

Table 2: Technical Design Specifications

TDS#	Reference	Metric	Range of Values	Notes/Motivation
1	R10 F1.2, F1.3	Static Load Capacity	$\geq 250kg$	The carts often weigh up to 200 kgs [3]. The device must be able to handle this weight, plus a safety factor.
2	R1 F2.1	Added Mass	$\leq 3.5kg$	The device must remain lightweight to prevent further fatigue to the operator.
3	R8 F2.4, F2.5	Actuation Force	$\leq 40N$	The force required to engage the device must be low to avoid repetitive strain injuries.
4	R4 F1.4	Maximum Overall Volume	300x300x150 <i>mm</i>	To remain discreet and avoid the device from interfering with the cart or the environment.
5	R9 F3.5	Operating Temperatures	$-5^{\circ}C$ to $45^{\circ}C$	To ensure that the device can survive the cold winter mornings and hot summer days without deformation or seizing [4].
6	R9, R11 F3.2, F3.3, F3.4	Water and Dust Protection	IP54	Protected from limited dust ingress and water spray from any direction [5] to ensure continuous operation
7	R2	Total Cost	$\leq R1000$	To remain affordable on the average waste picker's average income of R72.1 per day [6].
8	R6 F4.3	Part Replacement Time	$\leq 15 \text{ minutes}$	It must be possible to replace worn-out components (e.g., brakes or grips) quickly to reduce downtime.
9	R1	Maximum Pulling Height	1m	Must be comfortable to pull without strain [7]

5 Concept Generation

To improve the efficiency and quality of life for waste pickers in Pretoria, the aim is to reduce the physical strain and risk they endure when collecting waste. This section outlines concepts to reduce fatigue experienced by waste pickers when transporting their trolleys. The concept generation is divided into two sections. The first section will include concepts for braking and stability for the waste pickers' trolley. The second section covers the engagement mechanisms that will control the cart and its braking system. The final concept includes the selected braking and stability concept, as well as the selected engagement mechanism to ensure complete control of the trolley.

5.1 Braking and Stability Concepts

The braking and stability sections cover concepts 1.1 – 1.3; these concepts aim to introduce a system to an existing trolley to make it easier to control. The concepts include braking systems to control momentum when navigating gradients, and/or a system to reduce rolling resistance and vibrations.

5.1.1 Braking and Stability Baseline

Figure 4 shows the reference trolley that will be used as the baseline for the braking and stability concept development. This trolley is likely the equipment used by waste pickers. The trolley consists of a base frame that is supported by four caster wheels. Any solution proposed in this section must be compatible with this trolley configuration and provide a meaningful improvement in current performance.

The concepts will be evaluated against this baseline to determine whether they improve the existing trolley operation method. This comparison will help to identify which of the proposed concepts is most suitable for further development. Consideration will be given to braking effectiveness, stability, and usability.



Figure 4: Reference trolley [8]

5.1.2 Friction brake (Concept 1.1)

Concept 1.1 is a friction brake, as shown in Figure 5. This friction brake relies on a high-friction pad that is pressed against the road or sidewalk to slow the trolley and keep it stationary when unattended. The friction brake is a simple, low-cost solution that will give the waste picker some control over their trolley when navigating gradients.

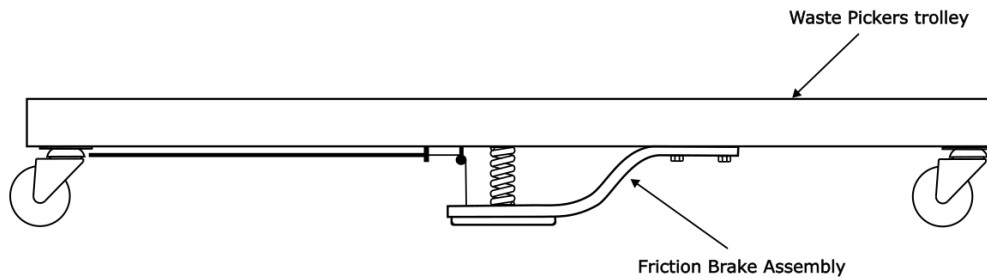


Figure 5: Side view of Concept 1.1

Figure 6 shows a close-up of the friction brake concept, providing a clearer view of its main components. The braking mechanism uses a compression spring and a spring plate to press a friction pad against the ground. When the friction pad is in contact with the ground, the friction force opposes the trolley's motion. This will make it possible to slow down the trolley and help to keep it stopped when the brake is engaged.

In its default position, the friction pad will stay in contact with the ground. This fail-safe mechanism will enhance trolley control, reduce required effort, and improve operator safety. To release the brake, a universal brake cable is attached to the spring plate. When the cable is pulled, the spring plate lifts, raising the friction pad off the ground.

The braking system is positioned underneath the centre of the trolley to ensure that the braking force is applied symmetrically. This will reduce the likelihood of the trolley tipping or changing direction unexpectedly when the friction brake is engaged.

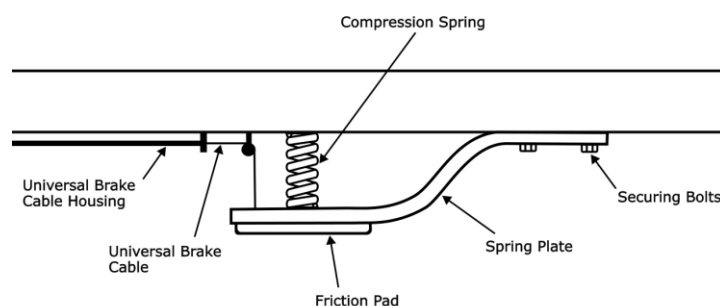


Figure 6: Close view of braking system for Concept 1.1

5.1.3 Rear Bicycle Wheel (Concept 1.2)

Concept 1.2 is a single wheel assembly mounted centrally at the rear of the waste pickers' trolley, replacing the rear caster wheels as shown in Figure 7. This concept improves the trollies' directional stability while also allowing for increased manoeuvrability when required. The rear-mounted wheel helps to guide the trolley in a straight path when the steering mechanism is locked.

During normal operation, the wheel will be kept in the straight position using a locking pin. This will prevent the wheel from rotating around its vertical steering axis, helping the trolley to maintain

its direction of travel. When greater manoeuvrability is required, the waste picker can remove the steering lock pin, allowing the wheel to rotate freely. This makes it easier to turn the trolley in confined spaces or when navigating around obstacles.

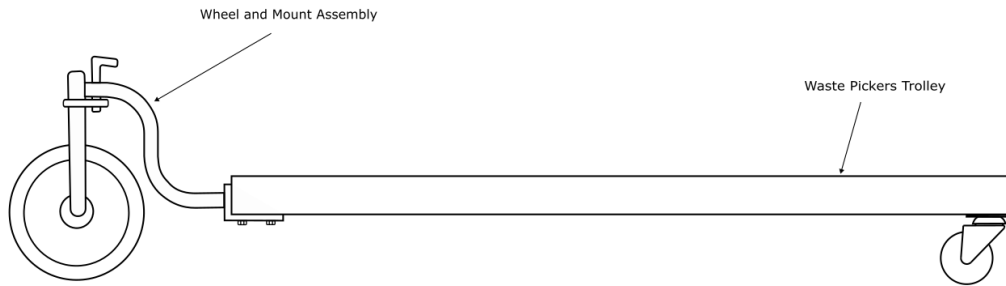


Figure 7: Side view of Concept 1.2

Figure 8 shows a close-up view of the rear wheel and mount assembly. The wheel is supported by a vertical fork, which is connected to the trolley's frame via a mounting bracket. The steering lock pin passes through the mounting bracket into a steering lock bracket attached to the bicycle fork, securing the wheel in the straight position.

This concept allows for a braking system to be integrated into the rear wheel assembly. Depending on limitations and wheel selection, either a rim (V-brake) or a hub brake can be selected. Including a braking system in the wheel assembly will provide more controlled braking than the original castor wheels. The lockable steering system allows the trolley to switch between improved straight-line stability and increased manoeuvrability at the waste picker's discretion.

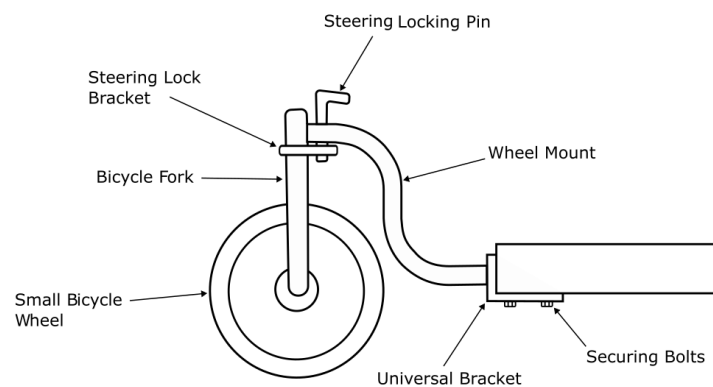


Figure 8: Close-up view of Wheel and Mount Assembly

5.1.4 Four bicycle wheels (Concept 1.3)

Concept 1.3 replaces the trolley's castor wheels with four small bicycle wheels. Figure 9 shows the side view of concept 1.3 and Figure 10 shows the front view. This concept aims to improve the straight-line tracking of the waste pickers' trolley by preventing the uncontrollable swivel that is likely to occur with castor wheels. By fixing the bicycle wheels to the trolley, the trolley is expected to travel more predictably in a straight line, improving overall control.

The use of bicycle wheels will likely also reduce the trolley's rolling resistance, especially when travelling over rough terrain. This will reduce the pulling force required of the operator and improve the trolley's overall usability when fully loaded. As the load will be distributed across all four corners, the concept will also provide stable support across the trolley.

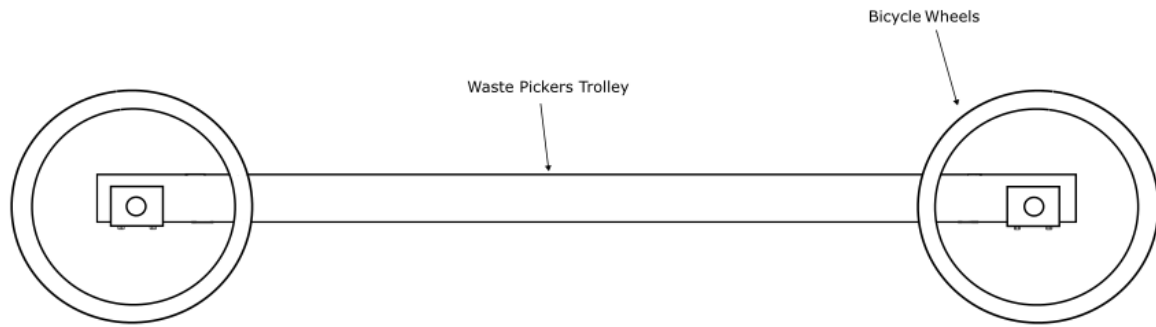


Figure 9: Side View of Concept 1.3

Each wheel is supported by a mounting bracket connected to the trolley's structure. The use of individual wheel mounts allows the wheels to replace the existing caster wheels of any trolley. This means there will be no loss of support compared to the reference trolley.

This concept will also allow the braking system to be integrated into one or more bicycle wheels. Depending on limitations and wheel selection, either a rim (V-brake) or a hub brake can be selected. Braking with multiple wheels will improve the trolley's stopping performance and provide greater control, especially when heavily loaded. Overall, concept 1.3 aims to improve straight-line stability, reduce the pulling force, and provide a more effective braking solution than the reference trolley.

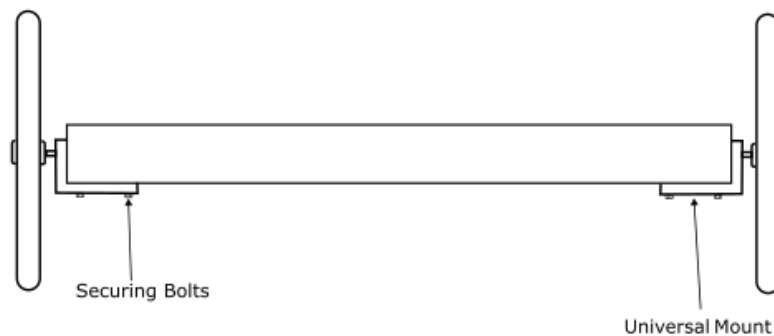


Figure 10: Front view of concept 1.3

5.2 Pulling Handle and Engagement Mechanisms Concepts

This section will focus on the concept generation of the pulling handle and engagement mechanism. These concepts aim to provide a comfortable and practical way for the waste picker to control the cart safely and effectively. Concepts 2.1 and 2.3 combine the pulling handle with the engagement mechanism. Each concept features a pulling bar that allows the waste picker to move the trolley while releasing the braking system without additional input. The concepts shown only include the reference trolley base, pulling handle, and braking mechanism.

5.2.1 Pulling Handle and Engagement Mechanism Baseline

Figure 11 (left) shows a typical pulling handle that may already be fitted to a waste picker's trolley. This handle accommodates the manual brake lever with a holding brake, as shown in Figure 11 (right). Most waste picker trolleys lack a braking system; therefore, a manual brake lever has been selected as a reference. This type of brake lever is commonly used to actuate bicycle rim or hub brakes and will therefore provide a simple and proven baseline for comparison. The proposed pulling handle and braking mechanism concepts will be evaluated against the combined system that includes the existing pulling handle and manual brake lever.



Figure 11: Reference Pulling Handle (left), Brake with a Lock (right)

5.2.2 Rotating Pulling Bar Brake Release (Concept 2.1)

Concept 2.1 is a vertical pulling bar, as shown in Figure 12. When the waste picker pulls the vertical pulling bar, the handle rotates forward into a more suitable position for pulling the trolley. The forward rotation will release the braking system, allowing the trolley to move freely.

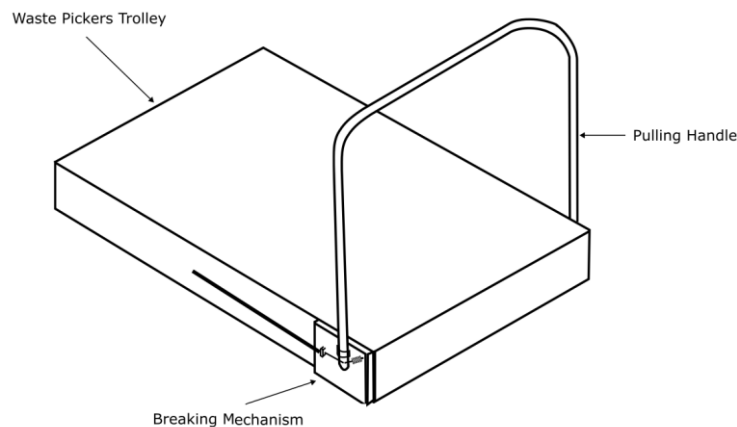


Figure 12: Isometric view of Concept 2.1

The braking mechanism is shown in Figure 13. As the pulling bar rotates forward, the tension in the universal brake cable is reduced, releasing the brake. When the force on the pulling handle is reduced, it will automatically rotate backwards into its original position. This returning motion is due to the tension spring that is attached to both the universal bracket and the pulling handle. As the pulling handle rotates backwards, the tension in the universal brake cable is reapplied, engaging the braking system and slowing and stopping the trolley.

The pulling bar in Concept 2.1 is connected to the trolley with a rotating bracket mounted to the universal bracket. This allows the handle to rotate relative to the trolley while also providing a mounting point for the tension spring and universal brake cable. Concept 2.1 provides an automatic braking response when the user releases or reduces force on the handle, but it requires the operator to maintain forward force on the handle while moving.

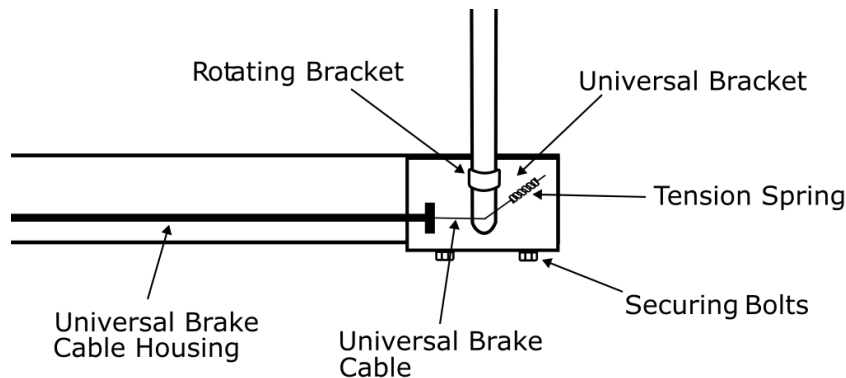


Figure 13: Close-up side view of the braking system for Concept 2.1

5.2.3 Sliding Pulling Bar Brake Release (Concept 2.2)

Concept 2.2 is a sliding pulling bar, as shown in Figure 14 and Figure 15. The pulling handle translates forward as the waste picker pulls the trolley. This releases the braking system, allowing the trolley to move freely. As the pulling handle does not move relative to the trolley, it remains in a constant, comfortable pulling position during operation.

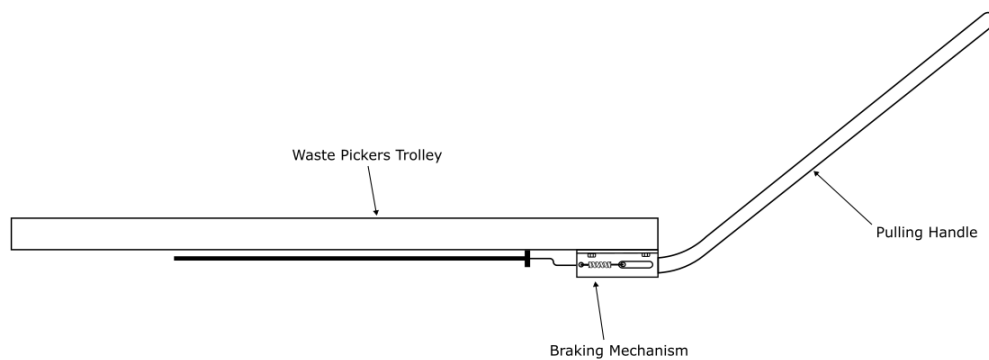


Figure 14: Side view of Concept 2.2

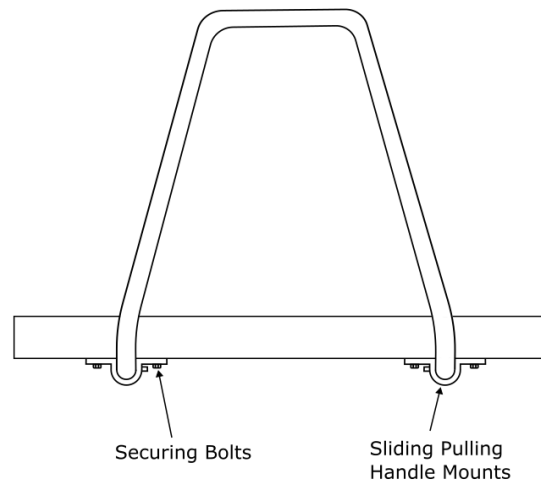


Figure 15: Front view of Concept 2.2

The braking function is shown in the close-up view in Figure 16. As the pulling handle moves forward, tension is applied to the brake cable, releasing the braking system. When the pulling force on the handle is reduced, the handle will slide backwards into its original position. This returning force is due to the tension spring connected to the handle mounts and the sliding mechanism. When the bar returns, the braking system will engage, slowing and stopping the trolley.

The pulling handle is guided and held in place by a sliding mount that allows for controlled linear movement of the handle to the trolley. A securing pin is attached to the handle, allowing for attachment of the tension spring and helping to limit motion. The sliding slot in the handle mount limits handle travel and prevents it from extending beyond the intended range. This concept provides automatic braking force as the pulling force on the handle decreases, while maintaining a constant pulling position.

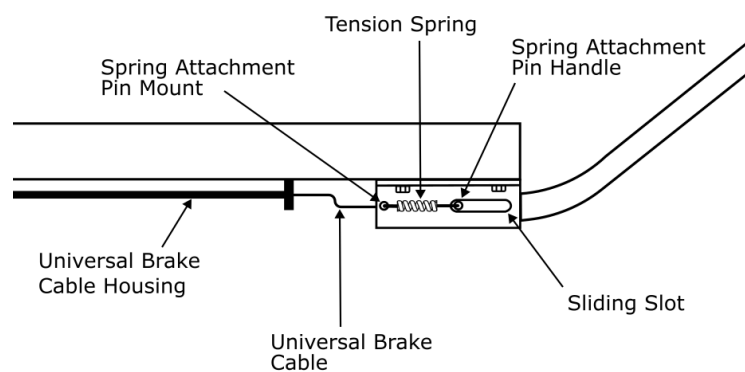


Figure 16: Close-up side view of the braking system for Concept 2.2

5.3 Concept Ranking

This section evaluates and ranks the concepts proposed in previous sections against one another and their respective baselines. Each concept will be assessed using relevant criteria and assigned a score of 1 to 5, where 1 indicates very poor performance and 5 indicates ideal performance. The coring system used for this evaluation is defined in Table 3.

The outcome of this section is to select a braking and stability system, as well as a pulling handle and engagement mechanism. The selected concepts will define the overall final design direction and provide a foundation for the detailed design of the individual components.

Table 3: Concept Evaluation System

Score	Meaning	Description
1	Very Poor	The concept performs poorly or not at all against the criteria and has major limitations.
2	Poor	The concept partially meets the criteria but shows clear weaknesses.
3	Acceptable	The concept meets the minimum requirements of the criteria, but does not show any clear advantages.
4	Good	The concept performs well against the criteria with minimal limitations.
5	Ideal	The concept fully satisfies the criteria.

5.3.1 Braking and Stability Ranking

This section compares and evaluates the proposed braking and stability concepts against the baseline trolley. Table 4 shows the main advantages and disadvantages of each concept. Table 5 defines the evaluation criteria that are used in the ranking process. Table 6 is the weighted ranking matrix, which is used to compare the concepts and identify the most suitable concept for further development.

Table 4: Advantages and Disadvantages of the Braking and Stability Concepts

Concept	Advantages	Disadvantages
Concept 1.1	- Simple mechanical system with minimal components. - Will require minimal maintenance. - It will be simple to install onto an existing cart. - No training will be needed for operation.	- Difficult to adapt to varying trolley heights due to the fixed spring-based braking force. - Reduced braking effectiveness on uneven or low-friction surfaces.
Concept 1.2	- Improved stability and manoeuvrability. - Reduced pulling force due to lower rolling resistance. - Braking performance is less affected by the ground surface. - Reduced vibration experienced by the operator.	- Requires operator training for correct operation. - Requires strong support due to the rear wheel offset. - Increased component costs may increase system cost.
Concept 1.3	- Minimal rolling resistance. - Improved load distribution across all four wheels. - Increased braking power and trolley control. - Improved operator safety during braking.	- Wheels may interfere with the collection bag. - Four bicycle wheels may increase costs. - Reduced manoeuvrability due to fixed wheel orientation.

Table 5: Criteria for Ranking the Braking and Stability Concepts

#	Criteria	Description	Weighting (%)
C1.1	Cost	The overall cost must remain realistic for waste pickers.	30
C1.2	Braking Effectiveness	The ability to slow the trolley, bring it to a stop, and keep it stopped.	20
C1.3	Directional Stability	The ability for the trolley to continue in a straight line without unwanted turning.	10
C1.4	Manoeuvrability	The ability for the trolley to steer and operate in confined spaces.	10
C1.5	Pulling Effort	The force required by the waste picker to move the trolley during normal operation.	10
C1.6	Ease of Integration	The ability to attach the concept to an existing cart.	10
C1.7	Durability	The concept's ability to withstand repeated loading, rough terrain, and outdoor use.	10

Table 6: Ranking for Braking and Stability Concepts

Criteria	Baseline	Concept 1.1	Concept 1.2	Concept 1.3
C1.1	5	4	3	2
C1.2	1	3	4	5
C1.3	1	1	4	5
C1.4	4	3	4	1
C1.5	1	1	4	5
C1.6	5	4	4	2
C1.7	3	3	4	3
Weighted Total	3.1	3.0	3.7	3.2

From Table 6, it is clear that Concepts 1.1 and 1.3 do not provide enough overall improvement when compared to the baseline. Concept 1.1 only improves the trolley's braking performance while increasing complexity and cost. This means it ranks below the baseline.

Concept 1.3 performs well on some criteria, particularly braking performance and straight-line stability. But it performs poorly in terms of manoeuvrability, cost, and ease of integration. Even though Concept 1.3 scores higher than the baseline, its limitations reduce its suitability for further development.

Concept 1.2 gives the most balanced improvement across the evaluated criteria, with an overall score approximately 20% higher than the baseline. It improves the braking performance, directional stability, and manoeuvrability while still remaining more practical than Concept 1.3. Therefore, Concept 1.2 is selected as the most suitable concept for further development.

5.3.2 Pulling Handle and Engagement Mechanisms Ranking

This section compares and evaluates the proposed braking and stability concepts against the baseline trolley. Table 7 shows the main advantages and disadvantages of each concept. Table 8 defines the evaluation criteria that are used in the ranking process. Table 9 is the weighted ranking matrix, which is used to compare the concepts and identify the most suitable concept for further development.

Table 7: Advantages and Disadvantages of the Pulling Handle and Engagement Mechanisms Concepts

Concept	Advantages	Disadvantages
Concept 2.1	- Braking control on downhill slopes. - Low maintenance requirements. - Simple mechanical operation. - Brake is applied when not in use	- Possible interference with the reclaimed goods. - Possible the handle will not be compatible with all carts. - Risk of incomplete brake engagement. - Variable pulling geometry. - Requires continuous force.
Concept 2.2	- Constant pulling geometry. - No interference with reclaimed goods. - Protected and secure mounting points. - Improved pulling comfort. - Brake is applied when not in use	- Mounts will require cleaning and lubrication. - Increased maintenance. - Possible braking system modifications will be required.

Table 8: Criteria for Ranking the Pulling Handle and Engagement Mechanisms Concepts

#	Criteria	Description	Weighting (%)
C2.1	Cost	The overall cost must remain realistic for waste pickers.	20
C2.2	Brake Engagement Reliability	The ability of the system to consistently engage the brakes when the pulling force is removed.	15
C2.3	Brake Release Control	The operator's ability to smoothly release and control the brakes while pulling the trolley.	15
C2.4	Ergonomics	Suitability of the handle position for comfortable pulling under normal operation.	10
C2.5	Interference With Cargo	The likelihood that the handle or braking mechanism might interfere with or be obstructed by the reclaimed goods.	10
C2.6	Mounting and Integration	The ability to mount the pulling handle to the trolley and operate it with the braking system.	10
C2.7	Maintenance	The expected need for lubrication, cleaning, and/or component replacement.	10

Table 9: Ranking for Pulling Handle and Engagement Mechanisms Concepts

Criteria	Baseline	Concept 2.1	Concept 2.2
C2.1	4	3	3
C2.2	4	4	4
C2.3	4	3	4
C2.4	3	5	3
C2.5	5	5	2
C2.6	3	3	2
C2.7	3	2	3
Weighted Total	3.4	3.2	2.8

It is clear from Table 9 that neither of the proposed automatic braking concepts provides meaningful improvement over the baseline system. An automatic braking system could improve and simplify the trolley's operation for the waste picker; however, the additional cost, complexity, and maintenance requirements make such systems less suitable for this intended use case.

Concept 2.1 has potential, as it will automatically engage the brakes when the pulling force is reduced. However, the rotating handle may interfere with the reclaimed goods and may not remain in a comfortable pulling position during operation. There is also the risk that, due to interference with the reclaimed good, the handle may not return to its original position, potentially reducing the braking system's reliability.

Concept 2.2 provides a more suitable arrangement for the pulling handle and braking mechanism. The handle maintains a constant geometry relative to the trolley, which improves pulling comfort and reduces the risk of fatigue. The mechanism is positioned so that it will not interfere with the reclaimed goods, and the mounting points are better protected from damage. However, the sliding brake release mechanism introduces additional complexity and maintenance requirements, primarily related to lubrication and cleaning.

The final design decision will adopt the pulling handle and mounting arrangements from Concept 2.2, but without the sliding brake-release mechanism. Instead, the mounts will act as fixed attachment points for the pulling handle. This will eliminate the need for lubrication and reduce maintenance associated with moving components. The manual brake lever from the baseline system will then be attached to the fixed pulling handle.

This configuration will combine the ergonomic and strength arrangement of Concept 2.2 with the proven manual brake lever from the baseline system. The waste picker will have a stable pulling handle that remains in a comfortable pulling position at all times, while maintaining full control over the braking system. The brake lever lock will keep the brake engaged when the trolley must remain stationary.

5.4 Final Concept

The final design that includes Concept 1.2 (Rear bicycle wheel) and the combination of Concept 2.2 and the baseline brake lever is shown in Figure 17. The brake lever is not shown in the figure, but it attaches to the top of the pulling handle near the operator's grip.



Figure 17: Final Concept Initial 3D Model

6 Detailed Design

This section presents the detailed design of the proposed improvements to the waste pickers' trolley. The focus will be on the wheel-mounting arm, universal attachment bracket, and the pulling handle. These components are shown in Figure 18. These components have been selected because they directly affect the trolleys' load capacity, ease of use, durability, manufacturability, and overall cost.

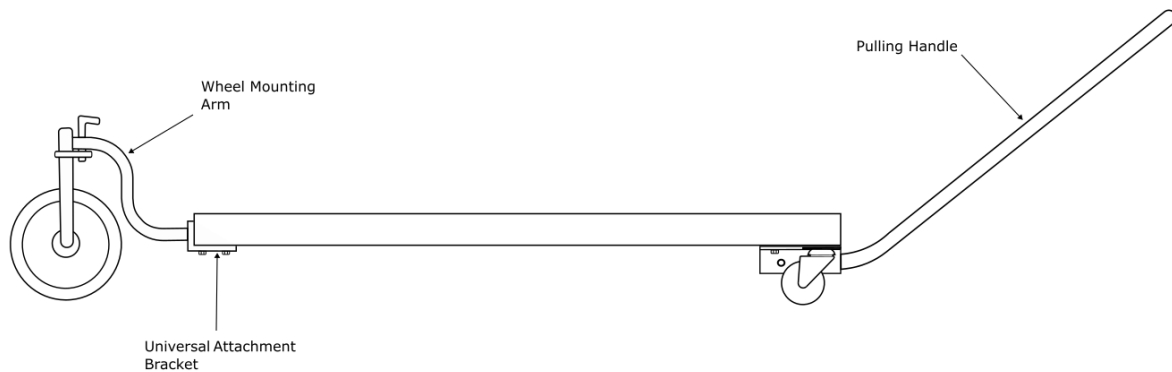


Figure 18: Components of focus for Detailed Design

The design approach is based on the following key aspects:

- **Safety and Reliability:** The trolley is going to be operating in uncontrolled public environments and may experience variable loading, misuse, impacts, uneven surfaces, corrosion, and long-term wear. For this reason, a global safety factor of 2 will be applied throughout the component design. This means that all components will be designed to a design load of 500kg, based on the maximum expected load of 250kg. The wheel-mounting arm and universal attachment will therefore be designed with strength and durability as the main requirements.
- **Ergonomics and Usability:** The pulling handle will primarily be designed around user comfort and ease of use during operation. Since this component is unlikely to experience excessive forces, the design can focus on enabling the waste picker to pull the trolley comfortably. This will help to reduce fatigue during operation.
- **Manufacturability and Cost:** The design must remain practical and affordable for production. Waste pickers will not benefit from a solution that is too expensive or too complicated to manufacture. This means the design uses standard components where possible and aims to keep the manufacturing process simple. This includes using bending rather than welding where suitable, maintaining constant bend radii, and limiting custom machining.

The rest of this section will discuss structural reasoning, geometric constraints, material selection, and stress-based justifications for the selected components. The wheel-mounting arm will be analysed first, as it is expected to carry the highest load and have the greatest influence on the trolley's overall structure.

The reference trolley used for all of the calculations is the same as the baseline trolley that is shown in Figure 4. The dimensions for this reference trolley are shown in Figure 19. This trolley has been selected as it is one of the larger trolley configurations available. This means that it represents a conservative case, as a smaller standard trolley would likely have a shorter moment arm, resulting in lower calculated stresses.

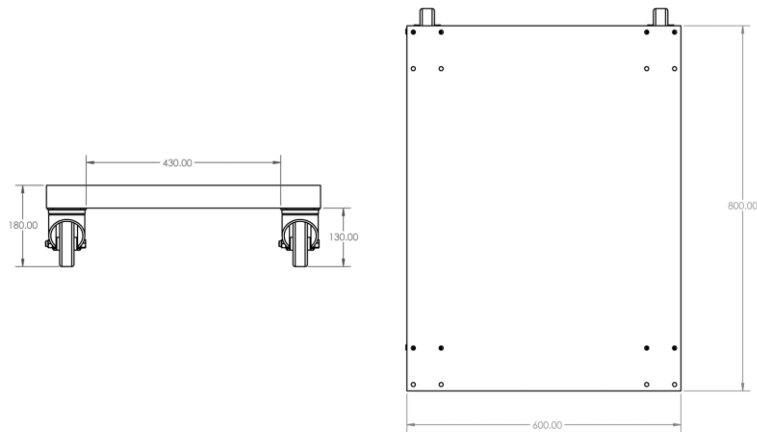


Figure 19: Dimensions of reference trolley

6.1 Wheel Mounting Arm

This section focuses on the wheel mounting arm. The wheel mounting arm extends from its connection point to the universal attachment bracket to the points where it connects to the steerer tube, which will be located at the top of the fork. The steerer tube will not be included in the calculations, but it will be cut from the same stock material to reduce costs and ensure compatibility in both geometry and material type.

The assumptions made at the beginning of this design process are intended to standardise the design and keep the components simple to manufacture, while maintaining strength. All bend angles are limited to 90 degrees with a constant bend radius. This will simplify the manufacturing process by providing a logical, standard bend angle that is easy to reproduce. The only welded connections will be with the universal attachment bracket and the steerer tube, as these connections are unavoidable. At both of these weld locations, the mounting arm will be positioned at a 90-degree angle to the relevant component. This avoids complex angled cuts in the tube and helps to keep the welded joints as simple and consistent as possible.

Some standard components must be defined early in the design process, as they play a significant role in the initial sizing of the wheel mounting arm and the trolley's overall dimensions. At this stage, the exact components will not be selected. However, a standard size is defined to specify the main dimensions.

Firstly, the bicycle wheel selected for this design is 12 inches, as shown in Figure 20 (right). This is the chosen size because it is the smallest commonly used wheel. A smaller wheel also reduces the amount of material used and the moment arm created, resulting in lower expected internal forces in the wheel-mounting arm. To accommodate the 12-inch wheel, a 12-inch bicycle fork is required, as shown in Figure 20 (left). The final standard component that is required at this stage will be a headset bearing, shown in Figure 21. This bearing has been selected as it has suitable dimensions to accommodate a 12-inch fork and is sealed to improve weather resistance.

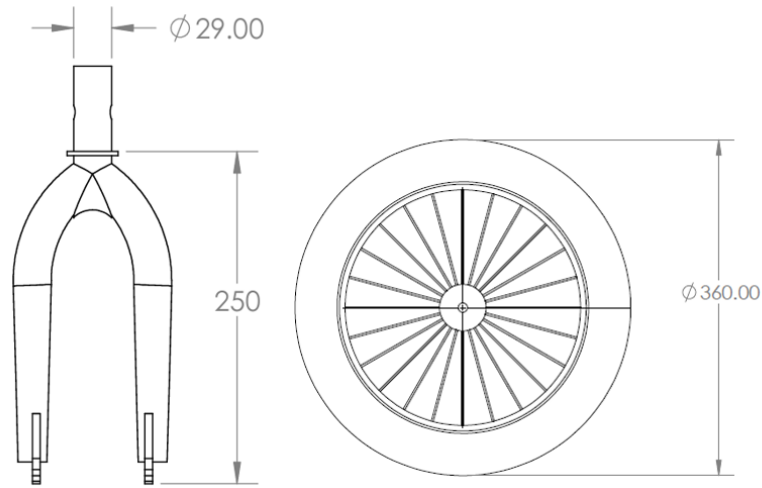


Figure 20: Dimensions of 12-inch Fork (left), 12-inch Wheel (right)

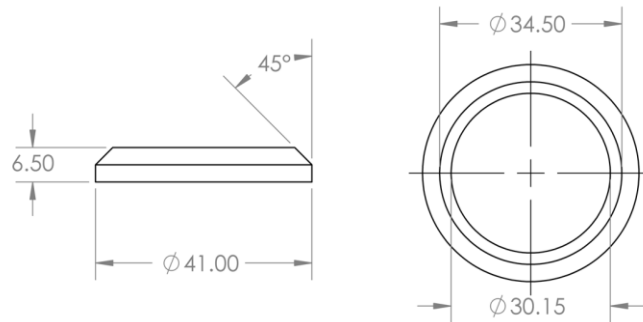


Figure 21: Dimensions of Headset Bearing

6.1.1 Initial Sizing of the Wheel Mounting Arm

The first step in sizing the wheel mounting arm is determined by the available standard stock size. As the steerer tube and the wheel mounting arm are made from the same stock material, the standard dimensions from the stock material are defined by the allowable size of the steerer tube.

Figure 22 (left) shows the isometric view of the steerer tube, while Figure 22 (right) shows the cross-section and corresponding dimensions. Where the ID is the inner diameter of the initial tube and t is the wall thickness. The top and the bottom of the tube will be flared out to accommodate the bearings. The inner diameter of the tube is the defining factor. If it is too large, it will not provide enough support to the headset bearings. If it is too small, the fork will not be able to fit. Therefore, the inner diameter of the tube must be greater than 29.00mm to allow sufficient space for the fork and less than 34.50mm to ensure adequate support for the headset bearings.

From the NJR Steel product catalogue [9], six possible stock sizes have been identified. The available dimensions that meet the required criteria are an Outer Diameter (OD) of 34.00mm with wall thicknesses of 1.6mm, 2.0mm, or 2.5mm, and an outer diameter of 38.10mm with wall thicknesses of 2.0mm, 2.5mm, or 3.0mm.

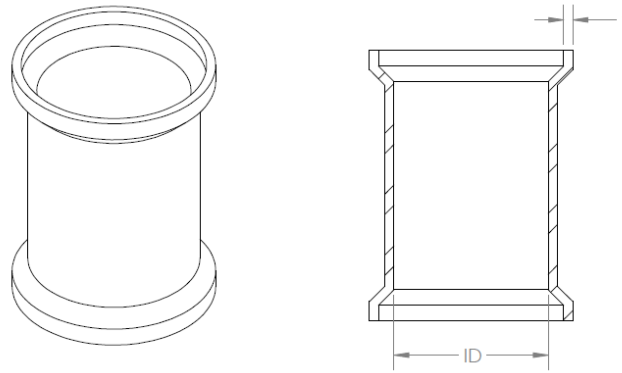


Figure 22: Steerer Tube Isometric view (left), Cross-section view (right)

Figure 23 shows the method for dimensioning the wheel mounting tube. Most of the dimensions relate to the tube's outer diameter. This is done to minimise strength loss during bending and to ensure that sufficient straight material is available to complete the bends properly.

The height of the vertical section (h) only needs to remain positive to ensure the bends are completed correctly. The centre height (H) is set to 300mm to ensure that the wheel remains at the correct height.

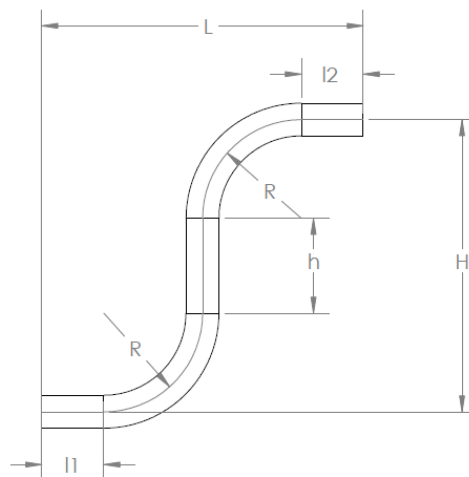


Figure 23: Sizing of the Wheel Mounting Tube

The horizontal lengths ($l1$ and $l2$), the bend radii (R), and the total horizontal length (L) are all related to the outer diameter of the tube. Table 10 shows their relation to the outer diameter of the tube, as well as the reasoning for each relationship.

Table 10: Wheel mounting tube dimensions with relation to OD

Dimension Symbol	Relation to OD	Reasoning
$l1$	$2(OD)$	To ensure that there is enough material to successfully bend the tube.
$l2$	$2(OD)$	To ensure that there is enough material to successfully bend the tube.
R	$3(OD)$	To ensure the bend radius is not too small and to avoid a reduction in material strength.
L	$10(OD)$	This dimension is defined by $l1$, $l2$, and R

6.1.2 Stresses in the Wheel Mounting Arm

For all example calculations in this section, a material stock with an outer diameter of 34.00mm and a thickness of 1.6mm will be used. The values for all of the available stock sizes will be included in tables in Appendix B.

To determine the stresses in the wheel mounting arm, the expected forces acting on the trolley and at the ground contact points must first be defined. The free-body diagram of the example trolley is shown in Figure 24. In this diagram, F_B represents the force that acts on the caster wheels, and F_F represents the force acting on the bicycle wheel.

The distributed load (q_L) represents the load from the reclaimed goods that the trolley is expected to carry. For the calculations, q_L will be represented as an equivalent point load (F_L) that acts at the centre of the trolley base (Point C). This load represents the design mass of 500kg, including the global safety factor.

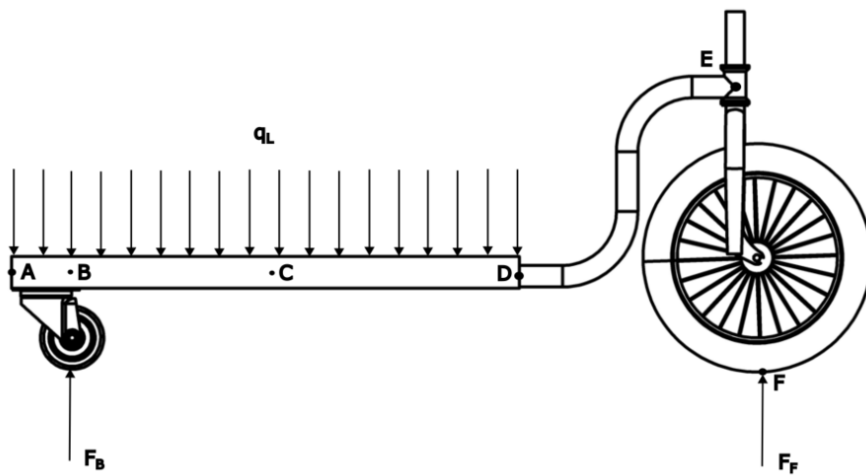


Figure 24: Free Body Diagram of the Example Trolley

The reaction forces at the ground contact points are determined using the equations of static equilibrium. The force acting at the caster wheels (F_B) is calculated by taking the moments about point F.

$$F_B = \frac{L_{CF}}{L_{BF}} F_L \quad (1)$$

Where L_{CF} is the horizontal distance between points C and F, and L_{BF} is the horizontal distance between points B and C.

The reaction force acting on the bicycle wheel (F_F) is calculated by summing the forces in the vertical direction.

$$F_F = F_L - F_B \quad (2)$$

The next step will be to isolate the wheel-mounting arm and determine the resulting forces acting on it. Figure 25 shows the internal forces and moments that act on the wheel-mounting arm. F_D and M_D represent the shear force and bending moment that act at the point where the arm connects to the trolley (Point D). F_E and M_E represent the shear force and bending moment that act at the point where the arm connects to the steerer tube (Point E).

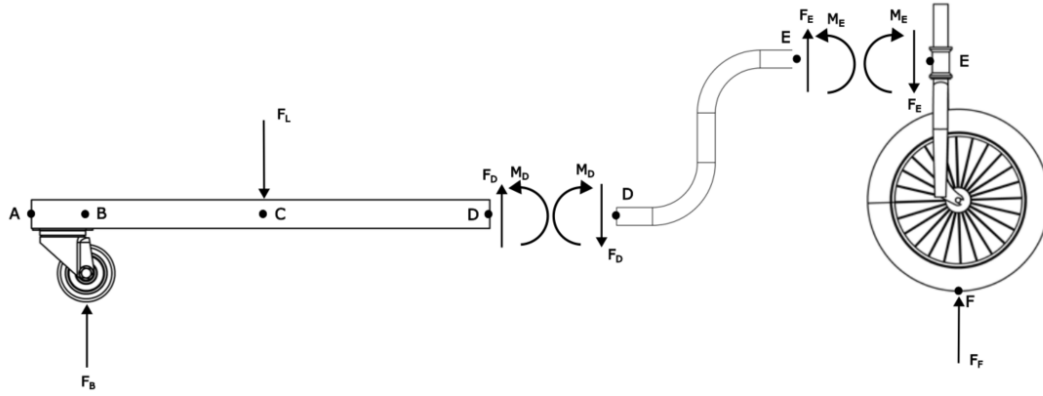


Figure 25: Internal Forces of the Example Trolley

The reaction forces at point D are determined using static equilibrium equations applied to the trolley base. The equilibrium equations for the reactions F_D and M_D are shown below:

$$F_D = F_L - F_B \quad (3)$$

$$M_D = L_{BD}F_B - L_{CD}F_L \quad (4)$$

Where L_{BD} is the horizontal distance between points B and D, and L_{CD} is the horizontal distance between points C and D.

The reaction forces at point E are determined using static equilibrium equations applied to the trolley base. The equilibrium equations for the reactions F_E and M_E are shown below:

$$F_E = F_F \quad (5)$$

$$M_E = L_{EF}F_F \quad (6)$$

Where L_{EF} is the horizontal distance between points E and F.

The results for equations 1-6 are shown in Table 11. These results are obtained for a stock with an outer diameter of 34.0mm and a thickness of 1.6mm.

Table 11: Results for equations 1-6 with a stock size of 34.0x1.6

Equation Number	Equation	Result	Unit
1	$F_B = \frac{L_{CF}}{L_{BF}} F_L$	3 516.54	N
2	$F_F = F_L - F_B$	1 388.46	N
3	$F_D = F_L - F_B$	1 388.46	N
4	$M_D = L_{BD}F_B - L_{CD}F_L$	520.67	Nm
5	$F_E = F_F$	1 388.46	N
6	$M_E = L_{EF}F_F$	48.60	Nm

Now that the reaction forces at either end of the wheel-mounting arm are known, work can begin on calculating the internal stresses in the arm. The most likely points of failure are at the connecting ends (Points D and E) and at the midpoints of each bend (b_1 and b_2). The internal forces are represented in Figure 26. N_1 , V_1 , and M_1 represent the normal force, shear force, and

bending moment at the midpoint of Bend 1. The same convention is used at the midpoint of Bend 2. Each bend midpoint is positioned at a horizontal distance of l_b from the nearest end of the wheel mounting arm.

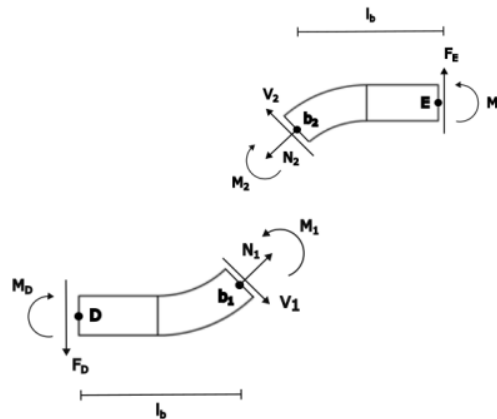


Figure 26: Internal Forces for Wheel Mounting Arm

As shown in Figure 26, the upper and lower sections of the arm will be analysed separately to determine the relevant internal forces. Table 12 shows the static equilibrium equations used, resulting expressions and the calculated results for the 34.00 x 1.6 stock size.

With the internal forces at all critical points known, it is now possible to determine the von Mises stress at each expected failure point. This will enable identification of the high-stress point along the arm and the definition of the material and final dimensions of the wheel-mounting arm.

Calculating the internal stress at the straight ends of the arm (Points D and E) is relatively simple. At these locations, the bending stresses will be calculated using straight-beam bending theory. The bending stress varies linearly across the section, with the maximum tensile stress on one edge and the equal maximum compressive stress on the opposite edge. The bending stress will be zero at the neutral axis, and this will be at the centre of the centroid.

This will not be as simple for the bend locations. At the bends, curved-beam theory must be used because the curvature of the member changes the stress distribution across its cross-section. Even though the stress still varies linearly throughout the cross-section, the tensile and compressive stresses are not mirrored across the centroidal axis. The zero-stress location will be shifted away from the centroidal axis by a distance of e .

Table 12: Equations and results for internal forces for the wheel mounting arm with a stock size of 34.0x1.6

Equilibrium Equation	Resulting Equation	Result	Unit	Equation Number
$\sum F_x = 0$	$V_1 = -\frac{F_D}{2 \cos(45)}$	-981.79	N	(7)
$\sum F_y = 0$	$N_1 = -V_1$	981.79	N	(8)
$\sum M_{b_1} = 0$	$M_1 = M_D - l_b F_D$	326.12	Nm	(9)
$\sum F_x = 0$	$V_2 = -\frac{F_E}{2 \cos(45)}$	-981.79	N	(10)
$\sum F_y = 0$	$N_2 = -V_2$	981.79	N	(11)
$\sum M_{b_2} = 0$	$M_2 = M_E - l_b F_E$	-243.15	Nm	(12)

The stress calculation method for the straight sections will be defined first, followed by the method for calculating stresses in the bend sections.

The stress calculations for both the straight and bent sections include some common section properties and stress equations. These include the cross-sectional area (A), the second moment of inertia (I), the distance from the centroidal axis (y), and the equation for calculating the von Mises stress ($\sigma_{von\ Mises}$). The equations for the equivalent von Mises stress are shown below:

$$\sigma_{von\ Mises} = \sqrt{\sigma^2 + 3\tau^2} \quad (13)$$

$$\sigma = \sigma_{normal} + \sigma_{bending} \quad (14)$$

For the straight sections, the shear stress (τ) and the resultant normal stress (σ) are found from the following equations:

$$\tau = \frac{F}{A} \quad (15)$$

$$\sigma = \frac{My}{I} \quad (16)$$

For the bent sections, the shear stress (τ) and the resultant normal stress (σ) are found from the following equations:

$$\tau = \frac{V}{A} \quad (17)$$

$$\sigma = \frac{My}{Ae(r_n - y)} + \frac{N}{A} \quad (18)$$

Where:

$$r_n = r_c - e \quad (19)$$

$$e = \frac{I}{r_c A} \quad (20)$$

$$r_c = 30D \quad (21)$$

Table 13 shows the resulting stresses at both edges of the tube for each of the critical points on the wheel mounting arm. The results clearly indicate that the most likely failure location is the connection between the arm and the trolley (Point D). Even if a reduction in the strength of the material is assumed in the bends, the von Mises stress at point D remains significantly higher than the maximum von Mises stresses calculated for either bend. Therefore, the stresses at point D are used as the reference for the material and dimension selection.

It is also clear that the bending stress is the dominant stress component. This indicates that when the trolley travels on a gradient, the peak stresses are likely to decrease as the resulting moment at point D is expected to decrease.

Table 13: Relevant stresses and combined von Mises stress for critical points for a stock size of 34.00x1.6

Critical Point	Side analysed	σ_{normal} (MPa)	$\sigma_{bending}$ (MPa)	σ (MPa)	τ (MPa)	$\sigma_{von\ Mises}$ (MPa)
D	Top (+y)	-	-413.18	-413.18	8.53	413.45
D	Bottom (-y)	-	413.18	413.18	8.53	413.45
E	Top (+y)	-	38.56	38.56	8.53	41.29
E	Bottom (-y)	-	-38.56	-38.56	8.53	41.29
b ₁	Inner radius (+y)	6.03	-286.99	-280.96	6.03	281.16
b ₁	Outer radius (-y)	6.03	238.65	244.68	6.03	244.90
b ₂	Inner radius (-y)	6.03	-213.98	-207.95	6.03	208.22
b ₂	Outer radius (+y)	6.03	177.94	183.97	6.03	184.26

Figure 27 shows a heatmap of the von Mises stresses calculated for all the available stock dimensions at point D. It is clear from this heatmap that the stock size of 34.00 x 1.6 produces the highest resulting stress. It may be possible to select a material that can withstand these stresses, but it may not be readily available and could increase overall cost.

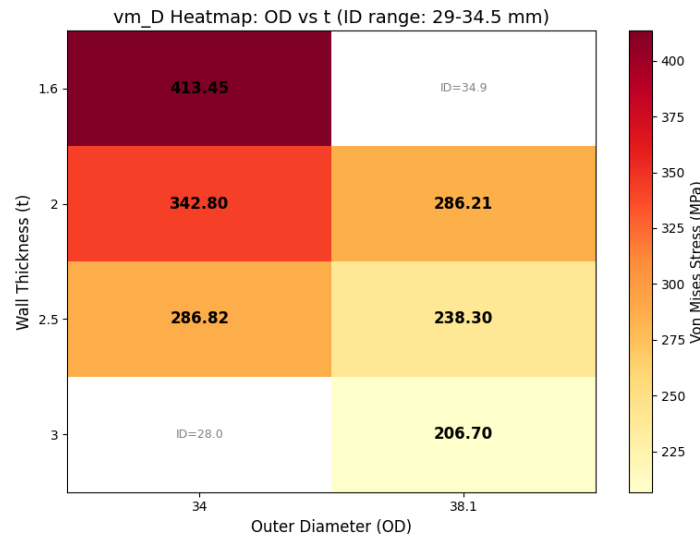


Figure 27: Heatmap for Von Mises Stress for each Available Dimension

The 34.00 x 2.5 stock size and the 38.10 x 2.0 stock size perform similarly in terms of stress and provide a good middle ground while keeping the dimensions to a minimum. Therefore, the next comparison will be based on the weights of the two components. A lighter wheel-mounting arm is preferred, as it can reduce overall costs and minimise additional mass on the trolley.

It was found that the 34.00 x 2.5 stock size has a volume of 136.67 cm³, while the 38.10 x 2.0 stock size has a volume of 132.21 cm³. This means that the 38.10 x 2.0 stock size is selected for the final stock dimensions for the wheel mounting arm.

6.1.3 Final Wheel Mounting Arm

To simplify the manufacturing process, the horizontal sections and bend radii have been rounded up to more practical values. The final design is shown in Figure 28. This design includes allowances for the steerer tube and the wheel locking pin.

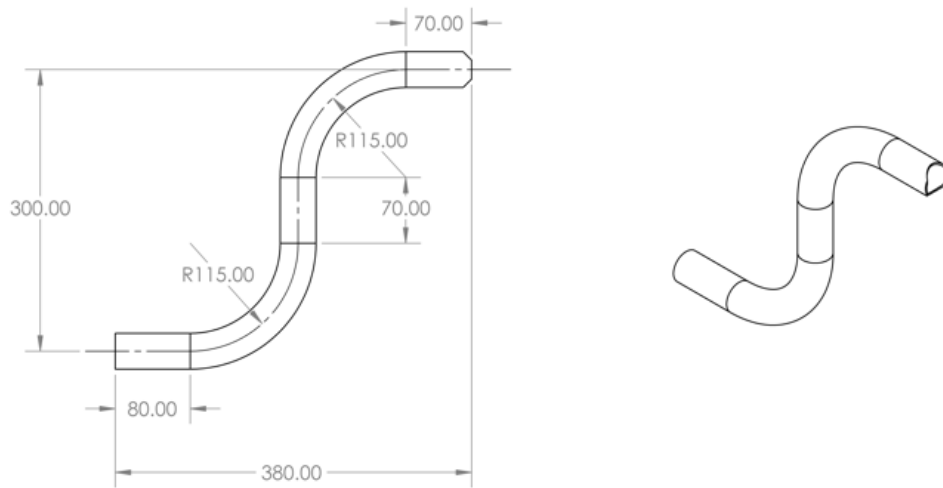


Figure 28: Final Wheel Mounting Arm

The final step in this process is to verify that the design satisfies the requirements and select a material for the wheel mounting arm. The material must withstand the calculated stresses while meeting the manufacturing and operational requirements. The detailed mechanical drawing is shown in Appendix C.

As the final design dimension has increased to improve manufacturability, the von Mises stress at point D has risen to 289.47 MPa. The selected material must be considered for low-temperature performance and ductility. The material must be able to safely operate at -5°C and must have a percentage elongation greater than 16.5% to ensure that the bends can be formed properly.

The selected material for this design is S355J2 carbon steel. This is a readily available material that also satisfies the main design requirements. It has a yield strength of 355 MPa, exceeding the required stress of 289.47 MPa, resulting in a final safety factor of 2.45. It is also tested at a temperature of -20°C and has a percentage elongation of 17%, satisfying the required 16.5%. This material is also well-suited for welding.

The main drawback of S355J2 is that it is not naturally water-resistant. Therefore, protective measures such as powder coating or painting will be required to reduce the risk of corrosion. Table 14 shows the final specifications for the designed wheel mounting arm.

Table 14: Final Wheel Mounting Arm: Important Properties

Property	Value	Unit
Safety Factor	2.45	-
Outer Diameter	38.10	mm
Wall Thickness	2.00	mm
Total Stock Length	591.28	mm
Total Stock Mass	1052.82	g

6.2 Universal Attachment Bracket

This section focuses on the bracket used to attach the wheel-mounting arm to an existing trolley. The main objective of this section is to develop a solution that securely attaches the wheel-mounting arm to the various trolley configurations used by waste pickers.

The proposed solution is an L-shaped bracket, shown in Figure 29. This attachment will be manufactured from two pieces of S355J2 steel plates. This material has been selected to ensure compatibility with the wheel mounting arm and to enable successful welding to it.

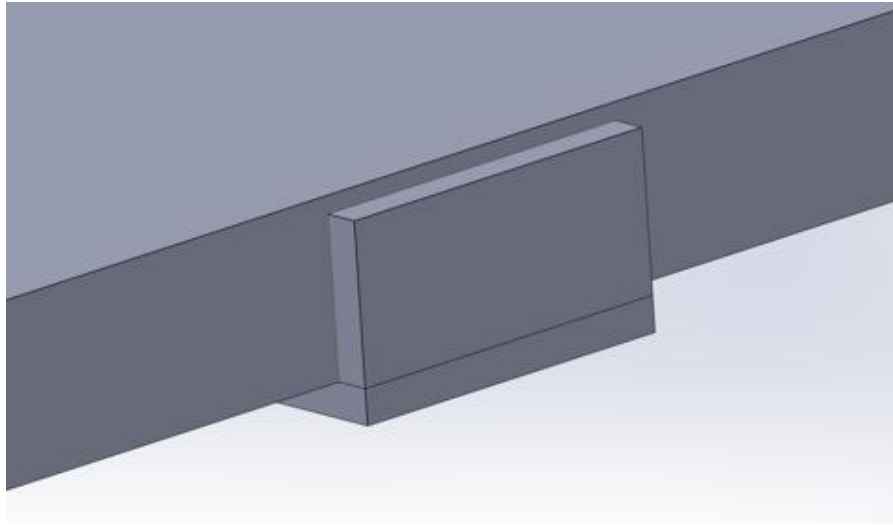


Figure 29: Initial Example of Attachment Bracket

To ensure the attachment bracket is compatible with a range of trolley configurations, it should not rely on the trolley's frame to hold the front plate in place. The force should rather be transferred to the trolley through the base plate. This will allow the load to be distributed more effectively across the trolley attachment area. By carefully considering the bolt attachment positions, it is possible to reduce areas of local high stress in the trolley's frame.

Figure 30 shows an example of the front plate that will be used for the attachment bracket. The height of the plate is fixed at 45mm to provide sufficient clearance for attaching the wheel mounting arm and for welding the front plate to the base plate. The width (w) and thickness (t) of the plate will be varied to determine a suitable combination of the two that can resist the forces and bending moment transferred from the wheel mounting arm.



Figure 30: Front Plate Example

Figure 31 shows an example of the base plate. The width (w) and the thickness (t) of this plate match those of the front plate. This is to ensure that the two parts are compatible and can be obtained from the same piece of stock material. This will help reduce costs by minimising the number of unique material sizes required.

A distance that is equal to t is left at the front of the base plate to ensure that there will be enough space to attach the front plate. The dimensions defining the bolt hole positions are measured from this point, as it marks the beginning of the trolley base. The front row of holes is positioned

at a distance of l_1 from the reference line, and the rear row of bolts is positioned at a distance of l_2 from the front row. The bolt holes are positioned as close as possible to the edges of the base plate, while leaving sufficient material around each hole. This will increase the spacing between bolt positions, helping distribute the force over a larger attachment area. The values of l_1 and l_2 are selected to reduce local stress concentrations at the bolt positions. The overall length (L) extends past l_2 to ensure sufficient material around each bolt hole while avoiding excess material.

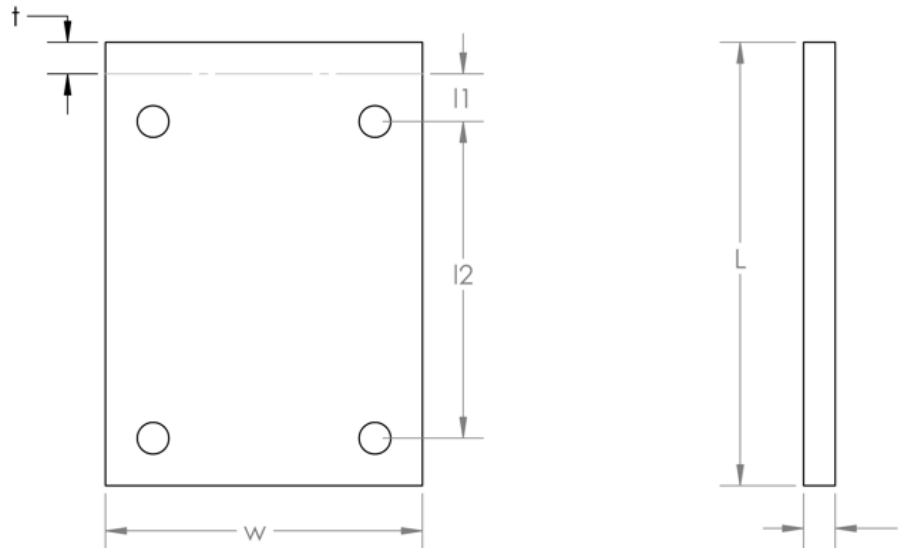


Figure 31: Base Plate Example

6.2.1 Front Plate

To determine the required width and thickness of the plates, the stress at the base of the front plate is calculated. A suitable combination of width and thickness will then be selected to ensure that the stress at the welded joint remains below the yield strength of 355 MPa for S355J2 carbon steel.

The free-body diagram of the front plate is shown in Figure 32. For this calculation, the front plate has been approximated as a cantilever beam. The loading on the plate consists of the bending moment transferred from the wheel mounting arm (M_D) and an axial force (F_D).

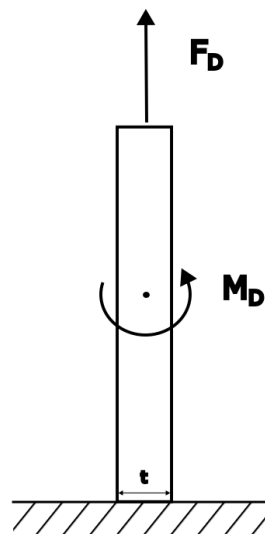


Figure 32: Free Body Diagram of Front Plate

The stress at the base of the front plate (σ_{base}) will result from both the stress from the bending moment (σ_{M_D}) as well as the stress from the axial force (σ_{F_D}). The equations that are used to calculate the stress at the base are shown below:

$$\sigma_{base} = \sigma_{M_D} + \sigma_{F_D} \quad (22)$$

$$\sigma_{M_D} = \frac{M \left(\frac{t}{2} \right)}{I} \quad (23)$$

$$\sigma_{F_D} = \frac{F_D}{A} \quad (24)$$

Where I is the second moment of inertia of the plate, and A is the cross-sectional area, they are found with the equations below:

$$I = \frac{wt^3}{12} \quad (25)$$

$$A = wt \quad (26)$$

From these equations, it is possible to determine the allowable plate thickness for a corresponding plate width. Figure 33 shows the allowable thickness required for different plate widths. Any combination of width and thickness that lies on or above the allowable thickness line will result in the stress at the base of the front plate being lower than the allowable limit of 355 MPa. If the selected combination lies below the line, the plate dimension will result in a stress that exceeds the allowable limit of 355 MPa. A suitable combination must be selected above this line while also keeping in mind manufacturing, material availability, and cost.

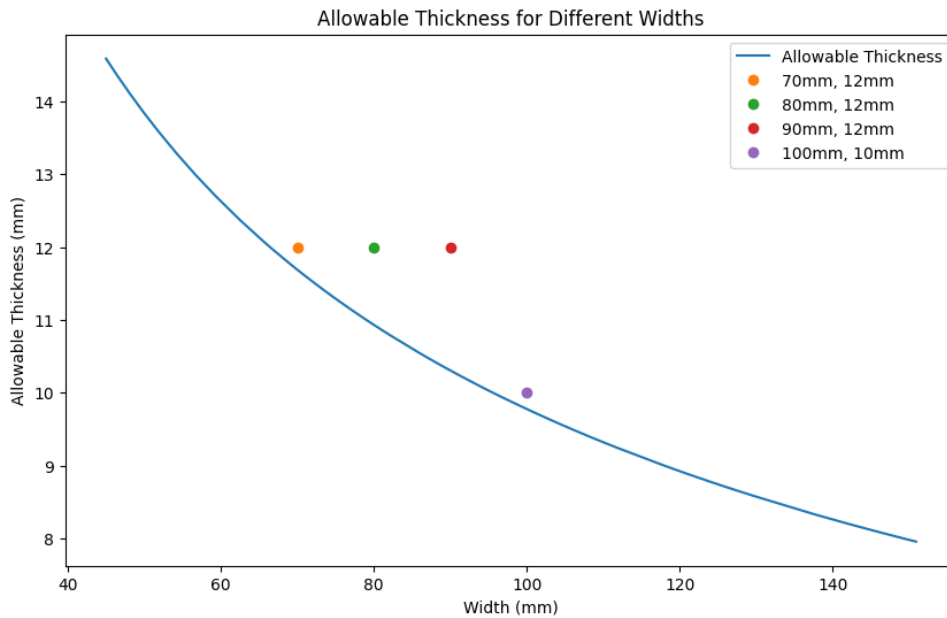


Figure 33: Graph Showing Allowable Thickness for Different Plate Widths

Figure 33 shows four combinations of plate dimensions that exceed the allowable thickness line. For each selected width, the plate thickness is set to the minimum available stock thickness that exceeds the allowable thickness. A plate width below 70 mm would result in a relatively thick plate, possibly increasing material usage and cost. A plate width above 100 mm is also less ideal,

as the width will become too large relative to the 38.1 mm diameter tube of the wheel-mounting arm.

Table 15 shows the selected combinations of plate width and thickness that satisfy the allowable stress requirements. The table also shows the corresponding stress at the base of the front plate and the mass for each option.

Table 15: Results for Acceptable Combinations

width, thickness (mm)	σ_{base} (MPa)	Mass (g)
70,12	336.8	296.73
80,12	294.71	339.12
90,12	261.96	381.51
100,10	339.24	353.25

The final selected combination is a width of 100mm and a thickness of 10mm. This option has been selected because its mass is not significantly higher than that of the lightest option, by only 56.52 g. The wider plate also makes the base plate wider. This allows the force transferred from the wheel mounting arm to be distributed over a larger contact area on the trolley base. Therefore, the selected dimensions provide a suitable balance between strength, mass, load distribution, and material usage.

6.2.2 Base Plate

Figure 34 shows the free-body diagram for the base plate. R_1 represents the reaction force for the back row of bolts, while R_2 represents the reaction force for the front row of bolts. The initial selection for the base plate design is to set l_1 to 15mm and use M10 bolts. These values will provide a good starting point for the design process and can be adjusted later if necessary.

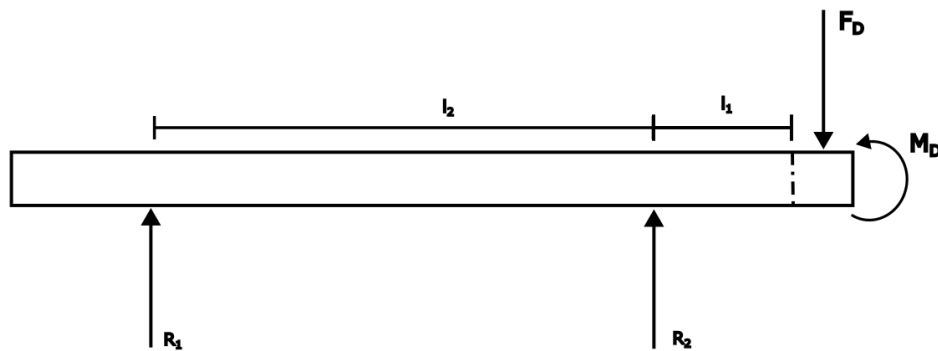


Figure 34: Base Plate Free Body Diagram

To determine the reaction forces (R_1 and R_2), the following equilibrium equations will be used:

$$R_1 = \frac{1}{l_2} \left[M_D - F_D \left(l_1 + \frac{t}{2} \right) \right] \quad (27)$$

$$R_2 = F_D - R_1 \quad (28)$$

For this arrangement of bolts, the front row acts as the pivot point. For this reason, the distance between the rows of bolts (l_2) will be selected based on the stress experienced in the back row of

bolts. The stress that is experienced in each of the bolts on the back row is calculated using the following equation, where the stress area (A_s) for an M10 bolt is 58.0 mm²[10].

$$\sigma_{bolt} = \frac{1}{2} \left(\frac{R_1}{A_s} \right) \quad (29)$$

Figure 35 shows the relationship between the tensile stress that is expected in each bolt in the back row and the distance between the bolts. From the graph, it is clear that M10 bolts provide more strength than is required. However, the M10 bolts will not be replaced to keep fasteners standardised and simple to replace, as M10 bolts are a common bolt size.

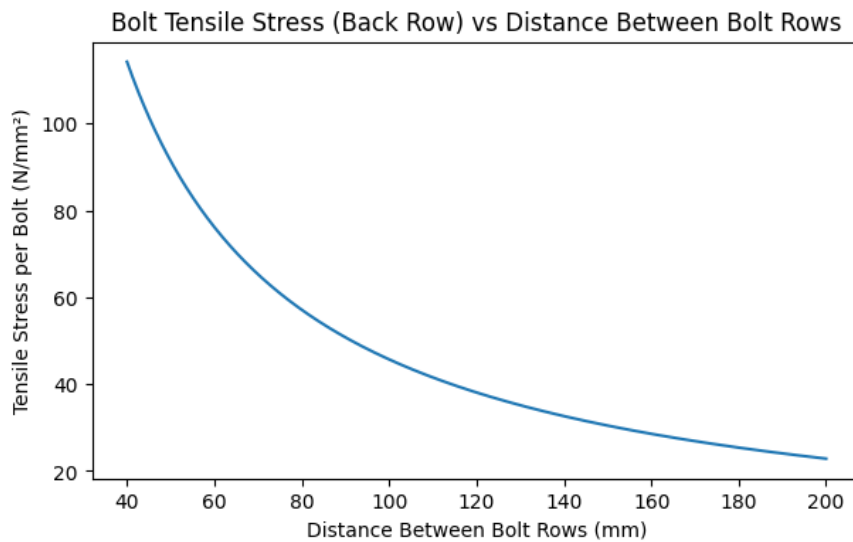


Figure 35: Graph Showing the Shear Stress of each Bolt on the Back Row

The distance l_2 is therefore selected to help distribute the compressive forces from the bolts, while also avoiding excessive material usage. A value of $l_2 = 100$ mm is selected. This provides a simple dimension for manufacturing and yields a tensile stress of 45.69 MPa per bolt. This is approximately 19% of the 240 MPa yield strength of a class 4.6 bolt [10]. This selection also ensures that if the waste picker is required to replace a bolt, even a lower-strength bolt, such as a class 4.6 bolt, would provide sufficient strength.

6.2.3 Final Universal Attachment Bracket

The detailed mechanical drawings for both the front plate and base plate are shown in Appendix C. The universal attachment bracket is shown in Figure 36. This bracket includes minor changes to simplify the manufacturing process and improve the mount's versatility.

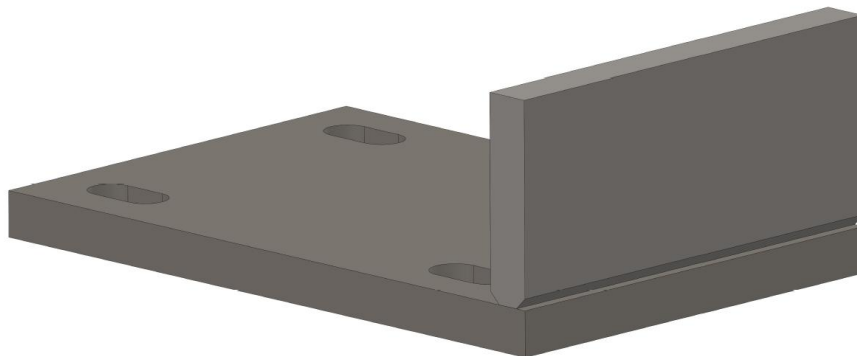


Figure 36: Final Universal Attachment Bracket

Firstly, chamfers are added to the bottom of the front plate. The chamfers provide extra space for the weld while also improving the weld penetration. They will reduce the length of the outward-extending weld leg, helping to minimise potential interference between the bracket and the trolley base.

The hole in the base plate has been extended into slots. The main dimensions remain unchanged, but the slots allow some adjustability when the bracket is attached to the trolley. This will increase the range of trolley configurations the mount is compatible with, including those with less regular geometry. The slots also provide increased tolerance during installation if the bracket and trolley are not perfectly aligned.

6.3 Pulling Handle

No force or strength calculations have been performed for the pulling handle, as it is not expected to experience excessive loads that could cause failure. Existing pulling handles used by waste pickers are often made of rope or an aluminium handle attached to the trolley with wire. The material chosen for this design is 6061 aluminium tubing with an outer diameter of 21.5 mm and a wall thickness of 1.6 mm. Aluminium has been selected as it is lightweight, low-cost, and corrosion-resistant. This is important because the constant friction from pulling, as well as the moisture from the waste picker's hands, may remove any protective coating applied to the pulling handle. The outer diameter of 21.5 mm was chosen as many V-brake levers are designed to be fitted to a tube of this size.

The dimensions of the handle were selected based on ergonomics and material usage. The pulling section is 200 mm wide to allow for both hands to be placed comfortably next to each other. The handle height is 880 mm, so the pulling section will be 1 m above the ground. This height will depend on the height of the trolley that the handle is attached to. The bottom sections are 110 mm long to allow for enough space for attachment, and they are also spaced 285 mm apart. This is the minimum distance required to allow for the 200 mm handle section. As the tubes are normally supplied in 6m stock lengths, the total tube length is limited to 3 m to allow 2 handles to be made from one stock length and to maximise material usage. The handle extends as far forward as possible to reduce the risk of the waste picker striking their heels while walking, resulting in a forward reach of 790mm. This will result in the pulling of the handle using the maximum available stock. These dimensions are shown in Figure 37.

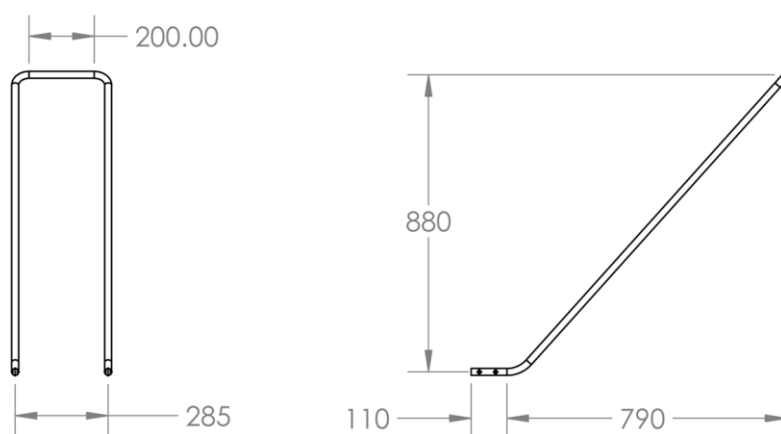


Figure 37: Pulling Handle Dimensions

The pulling handle is designed to allow the waste picker to operate the trolley in a more comfortable position while also having a suitable range of motion. This will help to reduce fatigue and lower the risk of injury during use.

7 System Integration

This section describes how the wheel-mounting arm and pulling handle are mounted on the trolley. The assembly discussion is followed by a detailed cost breakdown for each sub-assembly, along with the expected annual maintenance costs. The environmental, health and safety considerations of the final design are then discussed. This section concludes by comparing the final design with the technical design specifications defined earlier in the report. The final 3D model of each of the sub-assemblies attached to the reference trolley is shown in Figure 38.

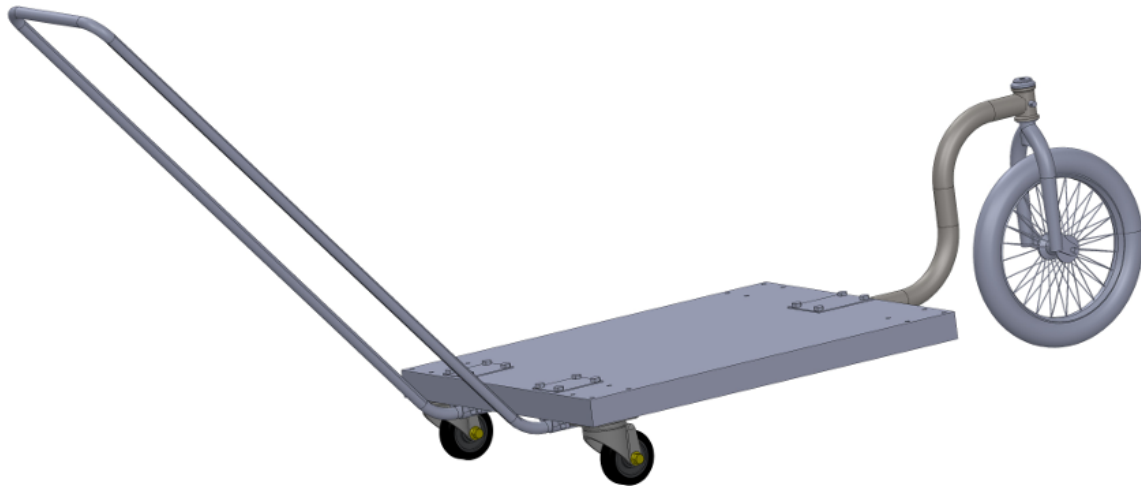


Figure 38: Final 3D Model

7.1 Assemblies

The wheel and mounting sub-assembly is discussed first, followed by the pulling handle sub-assembly. The method by which the sub-assemblies are intended to be attached to the trolley is also explained.

7.1.1 Wheel and Mounting

The wheel and mounting sub-assembly is shown in Figure 39. This sub-assembly includes the headset bearings that allow the wheel and fork to rotate freely when the bent locking pin has been removed, enabling the wheel to swivel for increased manoeuvrability. The detailed assembly is included in Appendix C along with the welding drawing for the wheel mounting arm.

The bicycle fork is secured to the wheel-mounting arm via a headset arrangement. A star nut is inserted into the top of the bicycle fork's steerer tube, providing a threading point for the M6x35 bolt. When the M6x35 bolt is tightened, the star nut will grip the inside of the fork's steerer tube, pulling it upward. This also pulls the top cap and compression ring downward, applying pressure to the top headset bearing. This brings all the headset components together and secures the fork to the wheel-mounting arm. This will secure the fork in place while still allowing the wheel to rotate with minimal unwanted movement.

A hole must be drilled through the steerer tube of the fork to allow the bent locking pin to pass through both the wheel mount and the fork's steerer tube. When the locking pin is inserted through the hole, the fork will be prevented from rotating. The bent lock pin will have a diameter of 10mm. This will allow replacement with an M10 bolt if the pin is lost or damaged.

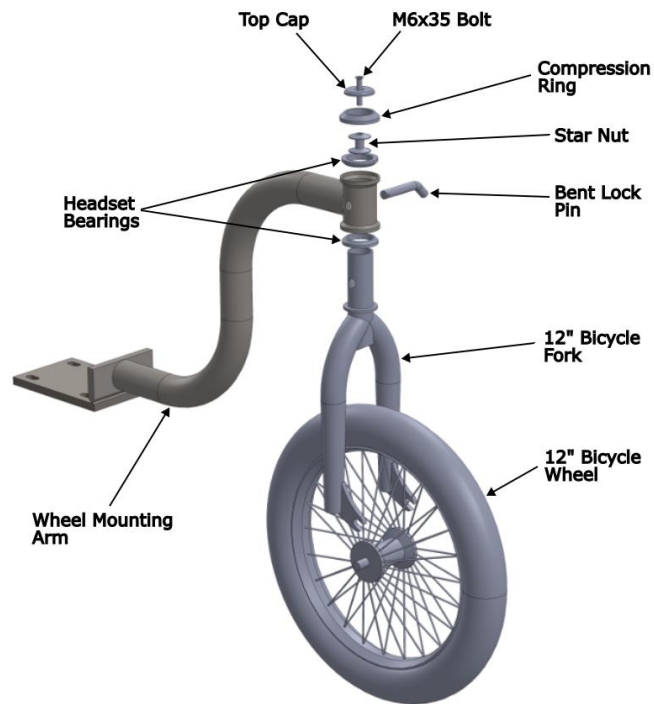


Figure 39: Wheel and Mount Sub-Assembly

7.1.2 Pulling Handle

Figure 40 shows a close-up view of the base for the pulling handle sub-assembly. An attachment bracket is bolted to each side of the pulling handle with two M10x50 bolts and M10 Nylock nuts. M10 bolts are used to maintain consistency with other fasteners used. Nylock nuts are used because they are designed to resist loosening under vibration, helping keep the bolts secure over long-term use.

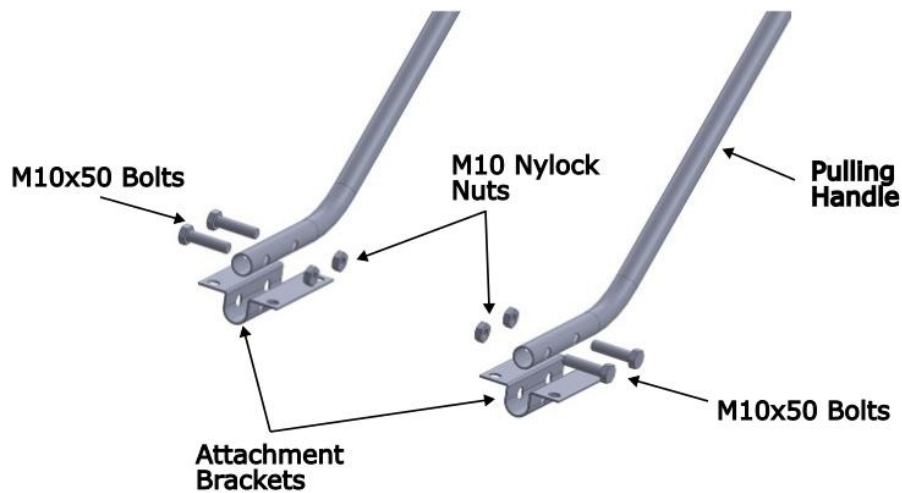


Figure 40: Close-up of the Pulling Handle Sub-Assembly

7.1.3 Total Assembly

Figure 41 shows how the sub-assemblies will be attached to the trolley. Before the improvements can be fitted, slight modifications will be required to the trolley base. These modifications should only include the drilling of the required holes for the attachment brackets.

The bicycle wheel is intended to replace the existing rear wheels on the trolley. This means the original wheels must be removed before the wheel-mounting assembly can be installed. Many trolleys include two swivel wheels and two fixed wheels. Where possible, the swivel caster wheels should be removed to improve the trolley's stability.

Each bracket is bolted to the trolley using four M10x70 bolts and M10 Nylock nuts. The bolt length is not fixed and can be selected to best suit each trolley. Distribution plates are used to spread the clamping force from the bolts over a larger area.

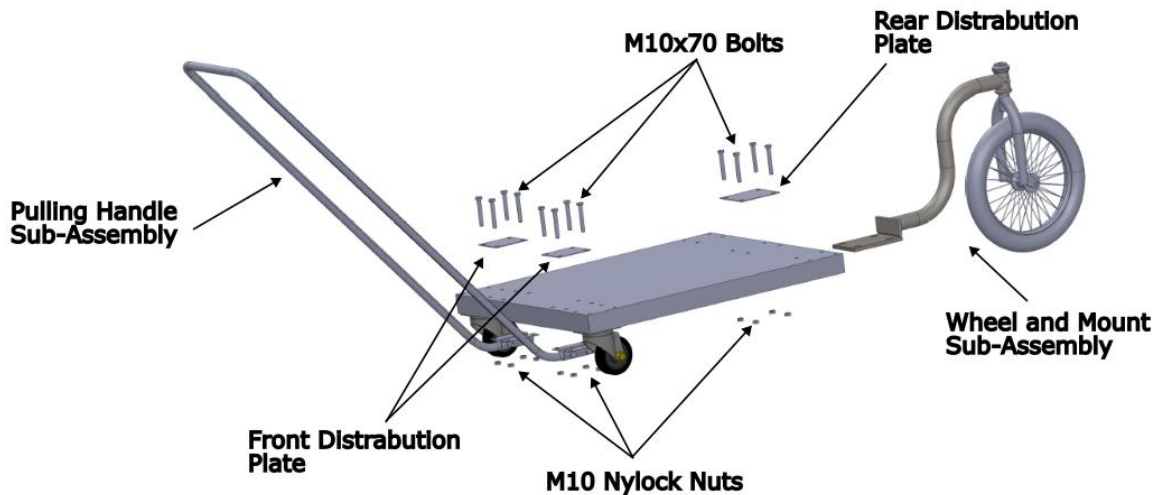


Figure 41: Attachment of Sub-Assemblies to the Trolley

7.2 Costs Analysis

This section will be split into 3 different sections: the wheel and mount, the pulling handle, and the braking system. The reason is that these sections are independent of each other, and the waste picker can use each one separately. The braking system will, however, require the waste picker to have the wheel and mount as well as a compatible pulling handle.

7.2.1 Wheel and Mounting Costs

Table 16 presents the cost breakdown for the wheel-and-mounting sub-assembly. It includes the cost of each component, the material and manufacturing costs where applicable, and the components needed to attach the sub-assembly to the trolley.

Table 16: Cost Breakdown for Wheel and Mount Sub-Assembly

Component	Manufacturing / Material Cost (if applicable)	Unit Cost	Quantity	Total Cost
12" Bicycle Fork	-	R160.42 [11]	1	R160.42
12" Bicycle Wheel	-	R140.00 – Rim [12] R73.98 – Tire [13]	1	R213.98
Headset bearing	-	R95.00 [14]	2	R190.00
Headset Top Cap + Star Bolt	-	R35.00 [15]	1	R35.00
Bent Pin + R-Clip	-	R65.00 [16]	1	R65.00
Wheel Mount	Manufacturing: R700.00 Material: R100	R800.00	1	R800.00
Distribution Plate	Laser Cutting: R100.00 Material: R20.00	R120.00	1	R120.00
M10x70 Bolt	-	R6.98 [17]	4	R27.92
M10 Nylock Nut	-	R0.74 [18]	4	R2.96
Total Cost				R1 615.28

7.2.2 Pulling Handle Costs

Table 17 presents the cost breakdown for the pulling handle sub-assembly. It includes the cost of each component, the material and manufacturing costs where applicable, and the components needed to attach the sub-assembly to the trolley.

Table 17: Cost Breakdown for Pulling Handle and Attachments

Component	Manufacturing / Material Cost (if applicable)	Unit Cost	Quantity	Total Cost
Pulling Handle	Manufacturing: R400.00 Material: R100	R500.00	1	R500.00
Distribution Plate	Laser Cutting: R100.00 Material: R20.00	R120.00	1	R120.00
M10x70 Bolt	-	R6.98 [17]	8	R55.84
M10x50 Bolt	-	R5.55 [19]	4	R22.20
M10 Nylock Nut	-	R0.74 [18]	12	R8.88
Total Cost				R706.92

7.2.3 Braking System Costs

Table 18 presents the cost breakdown for the braking system. It includes the cost of each component needed for proper fitment and operation.

Table 18: Cost Breakdown for Braking System

Component	Manufacturing / Material Cost (if applicable)	Unit Cost	Quantity	Total Cost
V-Break	-	R75.00 [20]	1	R75.00
Break Lever with Lock	-	R75.00 [21]	1	R75.00
3m Brake Cable Housing	-	R30.00 [22]	1	R30.00
3m Brake Cable	-	R85.00 [23]	1	R85.00
Total Cost				R265.00

7.2.4 Yearly Upkeep and Discussion

Table 19 presents the predicted yearly maintenance and upkeep costs for the components added to the trolley.

Table 19: Cost Breakdown for Upkeep

Component	Estimated Life (years)	Replacement Cost	Cost per Year
V-Brake Pads	3	R35.00	R11.67
Brake Cable Housing	1	R30.00	R30.00
Brake Cable	1	R85.00	R85.00
12" Tire	2	R73.98	R36.99
Headset Bearings	5	R190.00	R38.00
Total Cost			R201.66

The improvements are designed so that the waste picker can choose which additions will best suit their needs. It is also possible to purchase the improvements over an extended period rather than all at once.

From the available improvements, the bicycle wheel and mounting assembly is expected to provide the most benefit. This will reduce rolling resistance and the vibration that is experienced

during use. It also enables installation of the braking system, thereby improving the trolley's controllability and safety.

The braking system cannot be purchased or installed on its own, as it requires the wheel and mounting assembly. This is because the V-brake system requires a bicycle wheel and is not compatible with cater wheels. This means the combined cost of the wheel and mounting assembly, including the braking system, is R1 880.28.

The pulling handle is an optional improvement. If the waste picker already has a trolley that accommodates the brake lever, the pulling handle may not be required. In this case, the waste picker would not have to spend the R706.92. If the pulling handle is also required, the total cost of the improvement package will be R2 587.20.

This cost exceeds the specified target cost of R1000. However, approximately 50% of the total cost is attributable to the manufacturing of a few components. If the solution is implemented at scale, it may be possible to reduce the manufacturing cost and bring the total closer to the target cost.

The yearly maintenance cost is also relatively high. However, it is unlikely that the full maintenance cost will be incurred in any given year. The main exception to this would be when the headset bearings need to be replaced.

7.3 Environmental Impact Assessment

As the overall cost of the materials plays such a large role in the design, there will be minimal waste produced throughout the manufacturing process. All components made from sheet metal and metal plates will be strategically cut from the blank to minimise waste and material loss. All steel components are made from S355 steel, which is 100% recyclable. Due to the limited waste generated and efficient use of materials, the design's environmental impact is limited. As this design is intended to improve productivity in the informal recycling sector, the limited environmental impact of the proposed improvements may be offset by the increased volume of recycled goods collected.

The design is also focused on repairability, and the proposed improvements are also added to an existing trolley rather than replacing the full trolley structure. This may reduce the need to discard any damaged or incomplete trolleys. Individual components, such as the bracket, wheel, mounting arm or pulling handle, can be replaced if they are worn or damaged. This will help extend the usable life of each trolley.

This could ultimately contribute to higher recycling rates while also improving the quality of life for the waste picker using the trolley.

7.4 Health and Safety

From a health and safety perspective, the design aims to reduce the strain placed on the waste picker during trolley operation. The improved wheel arrangement helps reduce rolling resistance and vibration, resulting in less effort when operating the trolley, even on uneven surfaces. The pulling handle also improves the pulling position, allowing the trolley to be operated more comfortably, with better posture and control. The braking system improves the trolley's stopping ability, especially when travelling with a heavy load on a slope. The improved directional stability helps the operator maintain control of the trolley during movement. Overall, these improvements are intended to reduce fatigue, improve handling, and lower the risk of injury during daily use.

7.5 Performance Against Specifications

Table 20 compares the achieved values against the criteria specified earlier in the report in the technical design specifications. If a specification is not met, a reason is provided. In some cases, a possible solution is given to achieve the desired value or come closer to it.

At this stage, it is not possible to confirm whether all specifications have been met. Some of the values will need to be verified with testing. The table discusses whether the specification is expected to meet the value based on the current design.

Table 20: Comparison of Final Design Against TDS

TDS#	Metric	Target Values	Final Values	Status	Note/Comments
1	Static Load Capacity	$\geq 250kg$	500 kg	Met	All components have been designed with a static load of 500 kg. To meet a global safety factor of 2.
2	Added Mass	$\leq 3.5kg$	5 kg	Not Met	The total added mass slightly increased to accommodate a high load and ensure components do not fail
3	Actuation Force	$\leq 40N$	To be determined with testing	TBD	This value depends on the components used in the braking system, so it is not possible to provide a specific value.
4	Maximum Overall Volume	400x300x150 mm	390x290x150 mm	Met	Excluding the pulling handle, as it replaces an existing one. The wheel and mount fit within the target dimensions and do not interfere with operation.
5	Operating Temperatures	$-5^{\circ}C$ to $45^{\circ}C$	$-20^{\circ}C$ to $50^{\circ}C$	Met	The steel selected has been stress tested at $-20^{\circ}C$ and the plastics have a safe maximum temperature of $50^{\circ}C$.
6	Water and Dust Protection	IP54	IP54	Met	All components used are either sealed or have appropriate measures in place to ensure water resistance.
7	Total Cost	$\leq R400$	R2587.20	Not Met	Due to high component and manufacturing costs, it was not possible to meet the defined budget. It is possible to significantly reduce costs through bulk manufacturing. However, funding will be required for this solution to be viable.
8	Part Replacement Time	≤ 15 minutes	To be determined with testing	TBD	It is not possible to give a value at this stage; however, the design includes no complex parts, and all fasteners are constant. So the repair time will likely be below this defined time.
9	Maximum Pulling Height	$\leq 1m$	$\approx 1m$	Met	The pulling height will largely depend on the trolley it is attached to. The top of the pulling handle is designed to be 1m above the reference cart.

8 Conclusion and Recommendations

8.1 Conclusion

The main focus of this project was to improve the productivity, safety and usability of the waste pickers' trolley. The main aim was to reduce the physical strain experienced by the waste pickers while keeping the solution affordable, practical, and suitable for them.

The final design uses a rear bicycle wheel assembly, a universal attachment bracket, and a pulling handle with a manual braking system. The rear wheel is designed to reduce rolling resistance and vibration while improving directional stability without limiting manoeuvrability when needed. The universal attachment bracket allows the wheel mounting arm to be attached to a variety of different trolley configurations. The pulling handle allows the waste pickers to operate in a more comfortable pulling position. The addition of a braking system to the trolley improves the operator's ability to slow and control it, especially on steep gradients or with heavy loads.

The detailed design focused on the wheel-mounting arm and universal attachment bracket, as these components will carry most of the loads. A global safety factor of 2 was applied to account for variable loading, misuse, impacts and uneven surfaces. The selected design aimed to meet the required strength criteria while carefully considering manufacturability, material availability, and total cost.

The environmental and safety impacts of the proposed improvements were also considered. As the trolley's usability has improved, waste pickers can collect recyclable materials more effectively. This could improve recycling rates while also improving the working conditions and overall quality of life for the waste pickers who regularly use the trolley.

The main objective of the project, to develop a practical and affordable improvement that can reduce fatigue, improve control, and increase the usability of the trolleys, has ultimately been fulfilled. The design process throughout this project has provided a good opportunity to apply engineering principles to real-world problems that can have a meaningful impact on people's lives.

8.2 Recommendations

Testing should be conducted on various components of the solution before the design is implemented in practice. The wheel mounting arm, universal attachment bracket, bolted connections, and braking system should be tested with realistic loading conditions to confirm that all components perform as expected.

The components should be tested under different trolley configurations, on uneven surfaces, and over varying gradients. This will help to verify the compatibility, stability, and stopping ability of the trolley during normal operations.

Finally, the complete solution should be field tested with waste pickers to assess comfort levels, ease of use, and reusability. This will also result in feedback that can be used to refine the design and improve its suitability for the people who will be using it.

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Appendix A Meeting Log Card

Student name and surname:	Ethan Jackson
Student number:	u21481629
MOX410 topic:	Increasing productivity of waste pickers
MOX410 supervisor:	Dr WJ van den Bergh

Meeting	Date	Student signature	Supervisor signature	Supervisor comments
D1: Requirements analysis feedback	2026.02.25			Present.
D2: Functional analysis feedback	2026.03.04			Present.
D3: Concept design feedback	2026.03.18			Present.
D4: Engineering design feedback	2026.04.29			Present.
D5: System integration report	2026.05.20			Present.

Appendix B Calculation Results

Table 21: Ground contact forces for all available dimensions

OD (mm)	t (mm)	R (mm)	L (mm)	FB (N)	FF (N)
34.0	1.6	102.0	340.0	3516.54	1388.46
34.0	2.0	102.0	340.0	3516.54	1388.46
34.0	2.5	102.0	340.0	3516.54	1388.46
34.0	3.0	102.0	340.0	3516.54	1388.46
38.1	1.6	114.3	381.0	3567.27	1337.73
38.1	2.0	114.3	381.0	3567.27	1337.73
38.1	2.5	114.3	381.0	3567.27	1337.73
38.1	3.0	114.3	381.0	3567.27	1337.73

Table 22: Wheel mounting arm internal forces for all available dimensions

OD (mm)	t (mm)	FD (N)	MD (N-mm)	FE (N)	ME (N-mm)	V1 (N)	N1 (N)	M1 (N-mm)	V2 (N)	N2 (N)	M2 (N-mm)
34.0	1.6	1388.46	520674.14	1388.46	48596.25	-694.23	694.23	326115.72	-694.23	694.23	-243154.67
34.0	2.0	1388.46	520674.14	1388.46	48596.25	-694.23	694.23	326115.72	-694.23	694.23	-243154.67
34.0	2.5	1388.46	520674.14	1388.46	48596.25	-694.23	694.23	326115.72	-694.23	694.23	-243154.67
34.0	3.0	1388.46	520674.14	1388.46	48596.25	-694.23	694.23	326115.72	-694.23	694.23	-243154.67
38.1	1.6	1337.73	556494.55	1337.73	46820.45	-668.86	668.86	346441.53	-668.86	668.86	-256873.47
38.1	2.0	1337.73	556494.55	1337.73	46820.45	-668.86	668.86	346441.53	-668.86	668.86	-256873.47
38.1	2.5	1337.73	556494.55	1337.73	46820.45	-668.86	668.86	346441.53	-668.86	668.86	-256873.47
38.1	3.0	1337.73	556494.55	1337.73	46820.45	-668.86	668.86	346441.53	-668.86	668.86	-256873.47

Table 23: Wheel mounting arm von Mises stresses for all available dimensions

OD (mm)	t (mm)	D_inner (MPa)	D_outer (MPa)	B1_inner (MPa)	B1_outer (MPa)	B2_inner (MPa)	B2_outer (MPa)	E_inner (MPa)	E_outer (MPa)
34.0	1.6	413.45	413.45	281.16	244.90	208.22	184.26	41.29	41.29
34.0	2.0	342.80	342.80	233.68	202.61	173.09	152.42	34.14	34.14
34.0	2.5	286.82	286.82	196.09	169.08	145.27	127.16	28.47	28.47
34.0	3.0	250.00	250.00	171.40	147.00	127.00	110.52	24.73	24.73
38.1	1.6	346.54	346.54	233.90	204.30	172.20	152.93	31.76	31.76
38.1	2.0	286.21	286.21	193.61	168.41	142.56	126.04	26.14	26.14
38.1	2.5	238.30	238.30	161.63	139.88	119.04	104.66	21.68	21.68
38.1	3.0	206.70	206.70	140.56	121.05	103.53	90.55	18.74	18.74

Appendix C Mechanical and Assembly Drawings

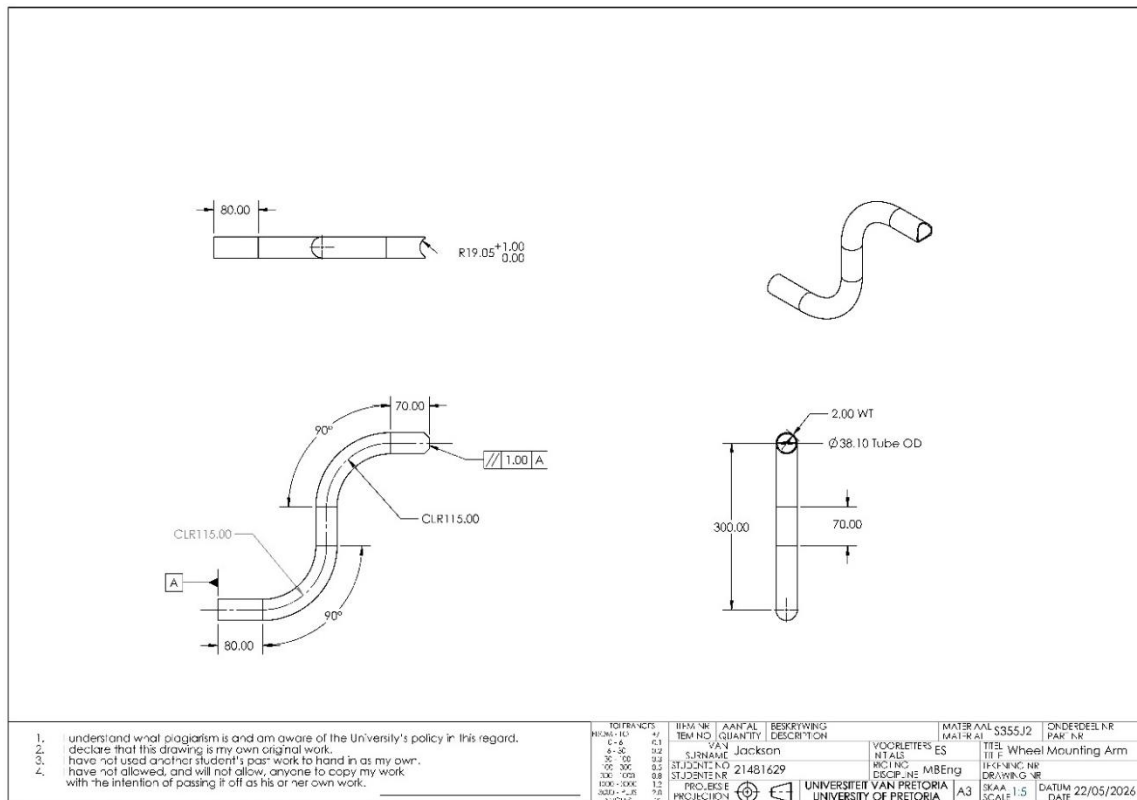


Figure 42: Wheel Mounting Arm Detailed Mechanical Drawing

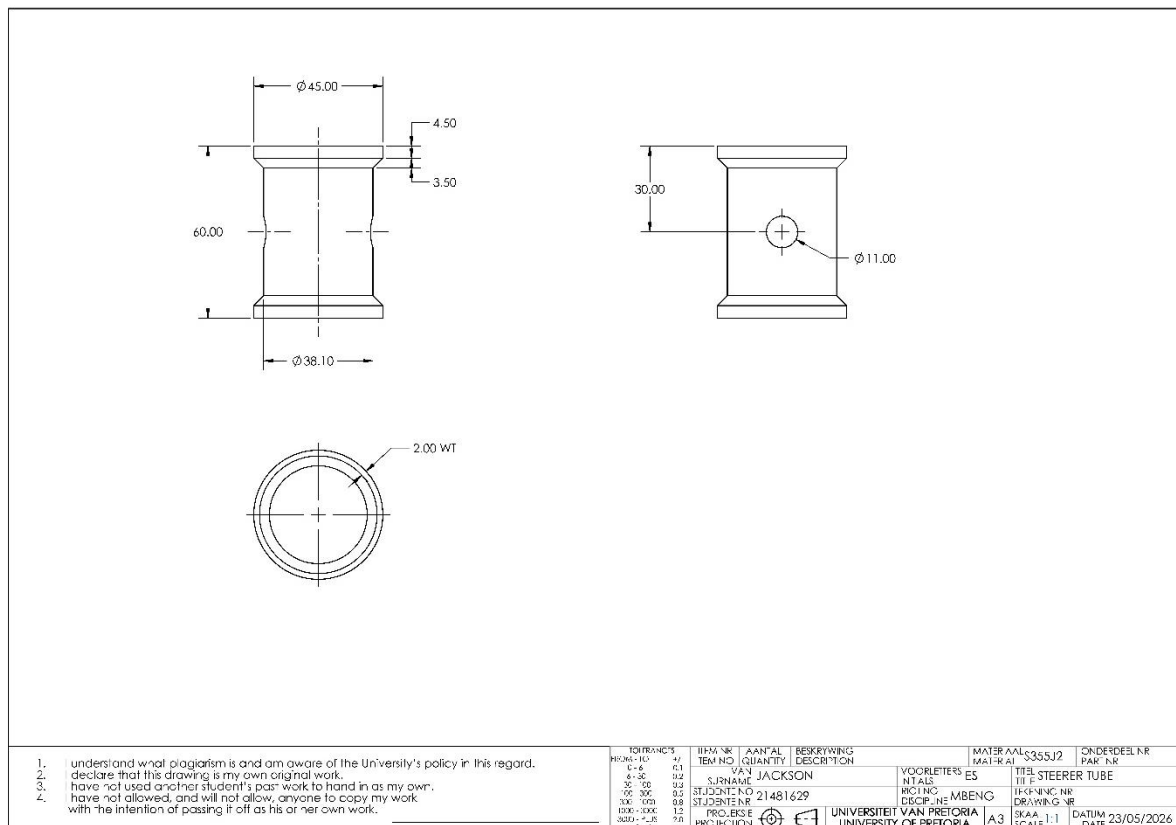


Figure 43: Steerer Tube Detailed Manufacturing Drawing

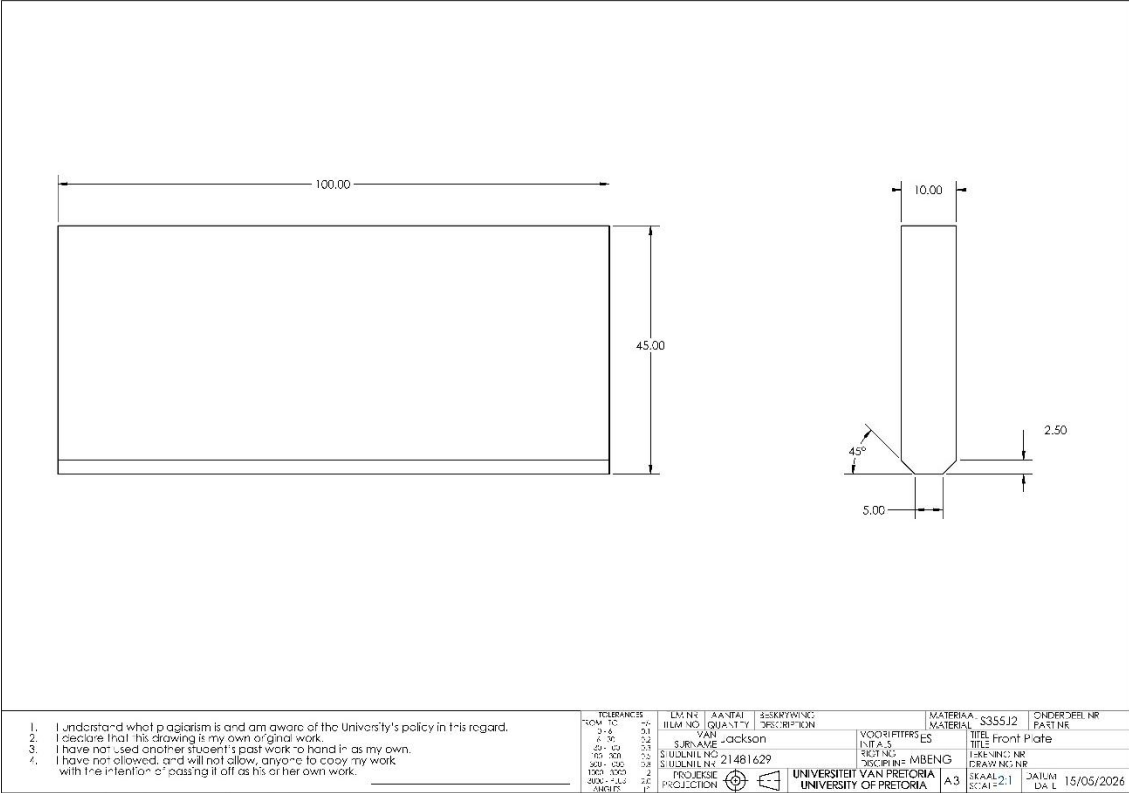


Figure 44: Universal Attachment Bracket Front Plate Detailed Mechanical Drawing

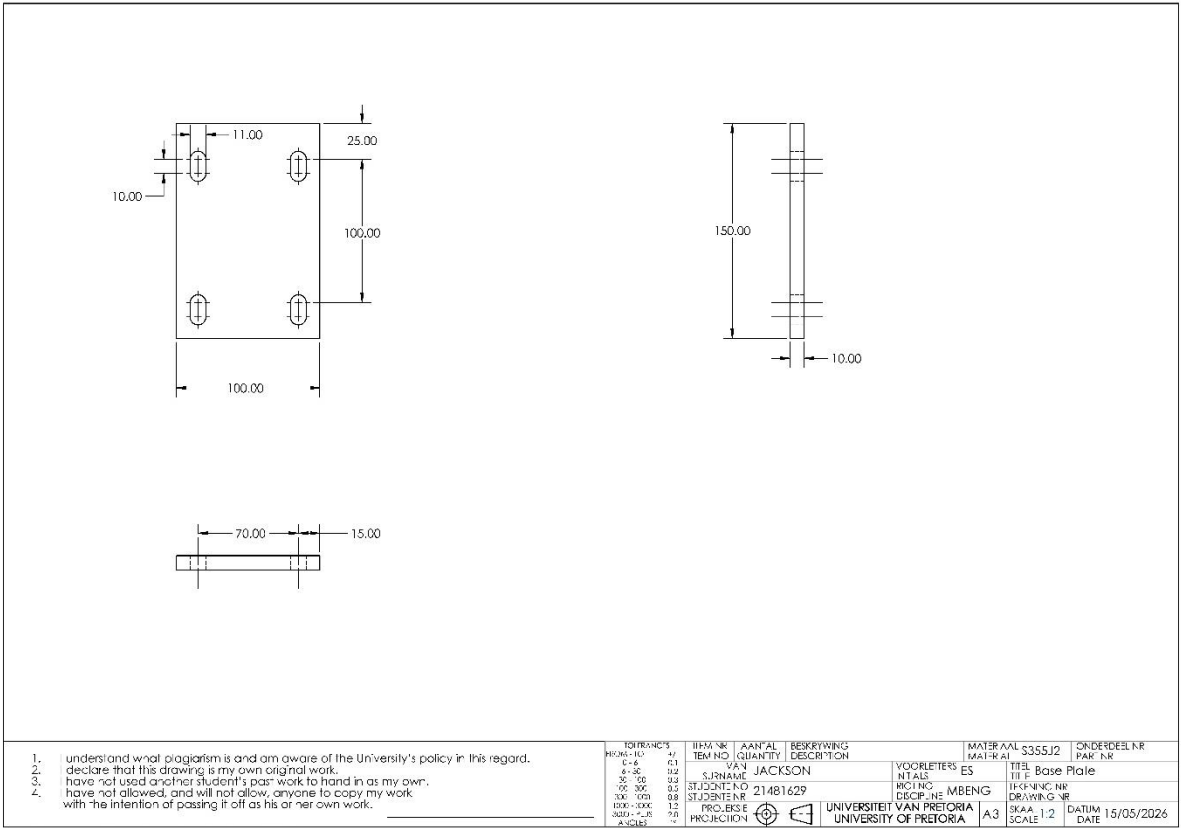


Figure 45: Universal Attachment Bracket Base Plate Detailed Mechanical Drawing

